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1-5 Outline of the

Chapter 2 Math

2-1 Introduction

2-2 Transfer Fun

2-3 Automatic C

2-4 Modeling in s

2-5 State-Space M
Equation Sys

2-6 Transformati

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5-3	Second-Order Systems
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	Problems

Chapter 8 PID Control

8-1	Introduction
8-2	Ziegler-Nichols

Contents

10-3	Solving Problems
10-4	Design of Sequences
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10-6	Design of Regulators
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10-8	Quadratic Optimal Control
10-9	Robust Control
	Example Problems
	Problems

Contents

addition, many problems are included, except Chapter 1. There are also quiz problems.

If this book is used as a text, a large portion of the material will be covered. The abundance of examples and problems allows many possible questions. This is a study book for practice.

I would like to thank the following: Dr. J. Bell, Cornell University; Dr. J. Iowa State University; Dr. J. Associate Editor, Mathematical Association of America; and Dr. J. involved in this publication.

control is an important part of modern manufacturing systems. Temperature, pressure, humidity, and other environmental factors are familiar with theory and practice.

This book is intended for use in college or university. A mathematical background is not required. The chapters are presented separately.

Brief Review of

The first significant work in the history of control theory was the work of James Clerk Maxwell in 1868. The first significant work in the history of control theory was the work of James Clerk Maxwell in 1868. The first significant work in the history of control theory was the work of James Clerk Maxwell in 1868.

tered around robust
Modern control
systems. Modern co
the theory is based
stability is sensitiv
means that when t
system, the system
system by first sett
troller in such a w
range, the designed
principle is called r
response approach
complex.

applied to abstract, word system should the like, systems.

Disturbances.

of the output of a s
internal, while an
an input.

Feedback Contr

of disturbances, ten
reference input and
turbances are so spe
pensated for within



Oil under
pressure

Fuel →

Figure 1-1
Speed control
system.

to apply techniques to represent the dynamic behavior of a relatively simple set of equations.

Although it is common to consider each of the component groups as a separate system, systems significantly interact with each other.

Consider, as an example, the human body. Major groups such as the nervous system, muscles, and organs, per se, are not functional. Experiments, product development, and analysis of human systems require that the groups are interconnected.

Such a system may be considered as a set of components necessary to accomplish a task. The dynamic characteristics of the system are determined by the performance of such components in the context of the overall system accomplishment.

Section 1-2 / Example 1-1

or an additive factor

or in other forms.

Since the exact
estimate of $\Delta(s)$ and
scalar transfer func

where $\|W(s)\|_\infty$ is the
infinity norm of $W(s)$

Feedback Control

between the output and the reference input, and is used as a means of controlling the system. For example, in a temperature control system, the output is the temperature of the space being heated or cooled, and the reference input is the desired temperature. The feedback control system uses the difference between the output and the reference input to determine the amount of heating or cooling equipment to use, so that the temperature remains at a constant level.

Feedback control is a common concept in many nonengineering fields, such as economics, psychology, and management. In a feedback control system, the output is measured and compared with the reference input, and the difference is used to adjust the system. For example, in a physiological system, the output is the body temperature, and the reference input is the desired temperature. The feedback control system uses the difference between the output and the reference input to determine the amount of heating or cooling equipment to use, so that the temperature remains at a constant level.

Section 1-3 / Closed-Loop Control

It should be emphasized that the time and in which the closed-loop control system is subjected to and/or unpredictable disturbances, the output power rating of the system. The number of components in a corresponding open-loop system is generally higher in a closed-loop control system. Closed-loop control may be used to improve the performance of a system.

Most analyses are performed with closed-loop control. Disturbances exist on the system.

avoid the possibility of instability. Compensators are most often designed to respond to electrical signals because of their reliability, ease of construction, and low cost.

Performance Specifications The designer must specify the tasks. The requirements for the system are given in the performance specifications. The specifications are given in the form of requirements (such as rise time, settling time, overshoot, etc.) or steady-state requirements. The specifications must be given in frequency domain. The specifications must be given before the design is started.

For routine design, the specifications are given in the form of steady-state error, relative stability, and bandwidth. In other cases, the specifications are given in the form of rise time, settling time, overshoot, etc.

Section 1-4 / Design

carries, distributed
the original design v
differ from the theo
quirements on per
changes in the proto
she must analyze ea
the next trial. The d
ifications and, at the

1-5 OUTLINE OF THE BOOK

This text is organized
as follows:

Chapter 1 prese

technique. The frequency design method until recently was the primary design method until robust control for design is gaining popularity.

Chapter 8 discusses the freedom PID controllers have in integral gain, and derivative gain. Controllers used have been based on the relative magnitudes of the time constants and magnitudes of the transfer function.

Ziegler and Nichols (1942). Since then numerous variations of PID controllers have been developed. Optimization approaches have been used to tune PID controllers.

Section 1-5 / Outline

economic, biological,
Such differential equations
ular system—for example,
for electrical systems.
mathematical models is the

Throughout this book,
considered. This method
depends on the past history
(the input for $t > 0$).

Mathematical Models

Depending on the physical
cal model may be better
lems, it is advantageous

In an experiment proportional, thus implying that the system can be considered linear.

Linear Time-Invariant Systems
A differential equation is linear if the dependent variable and its derivatives appear in the equation in lumped-parameter form. Linear time-invariant equations—those whose coefficients are constant—are called *linear time-invariant* (LTI). They are represented by the transfer function. (The system is a spacecraft engine's fuel consumption.)

By using the concepts of algebraic equations, the transfer function is equal to

Comments on The transfer function is limited to linear systems. The transfer function approach is not applicable to nonlinear systems. In what follows, we will consider the transfer function approach. (Note that a system can be represented by a differential equation or a transfer function.)

Section 2-2 / Transfer Functions

where both $g(t)$ and

Impulse-Response
invariant system to
Laplace transform
output of the system

blocks. The *junction* on the input signal components are used to rows to indicate the in the direction of the a unilateral property.

Figure 2–1 shows the block indicates the output. Su

Figure 2–1
Element of a block
diagram.

verted to a force or
This conversion is ac
as shown in Figure 2
it is compared with t

Figure 2–3
Block diagram of a
closed-loop system.

eliminating $E(s)$ from

or

The transfer function
relates the closed-loop
and feedback elements

From Equation

Section 2-3 / Auto

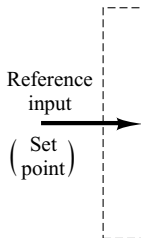
(b)

Figure 2–5

- (a) Cascaded system;
- (b) parallel system;
- (c) feedback (closed-loop) system.

(c)

Figure 2-6
Block diagram of an industrial control system, which consists of an automatic controller, an actuator, a plant, and a sensor (measuring element).



son, is very widely used.
Let the output signal be y .
In two-position control, the output is either high or low, depending on whether the error signal is positive or negative.

where U_1 and U_2 are the high and low output levels, respectively. Two-position control is widely used in the control of on-off valves. High gains act as two-position controllers.

Figures 2–7(a) and (b) show the response of a two-position controller. The range through which the output signal varies is called the range of the controller.

control system;
(b) electromagnetic
valve.

Figure 2-9

Level $h(t)$ -versus- t
curve for the system
shown in Figure 2-8(a).

where K_i is an adjus

Proportional-PI
plus-integral contro

where K_p is the proportional gain.
The block diagram of a proportional-plus-integral-plus-derivative controller is shown in Figure 2–10.

Figure 2–10

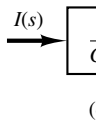
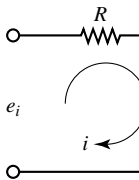
Block diagram of a
proportional-plus-
integral-plus-
derivative controller.

Consider now the case, the closed-loop of the disturbance is

On the other hand as the gain of $G_1(s)$ the closed-loop transfer and inversely proportional affect the closed-loop closed-loop system. $H(s) = 1$, tends to

Figure 2-12

- (a) RC circuit;
- (b) block diagram representing Equation (2-6);
- (c) block diagram representing Equation (2-7);
- (d) block diagram of the RC circuit.



(c) $\xrightarrow{R}$ ()

(d) $\xrightarrow{R}$

(e) $\xrightarrow{R}$

Figure 2–13

(a) Multiple-loop system;
(b)–(e) successive reductions of the block diagram shown in (a).

tially time-domain (e.g., H-infinity control),
approach. Before we
and state space.

State. The state
variables) such that
the input for $t \geq t_0$
 $t \geq t_0$.

Note that the co
cable to biological s

State Variables
ing up the smallest s

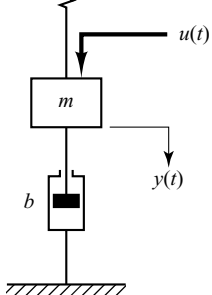
Section 2-4 / Mode

Assume that a m
also that there are
Define n outputs o
system may be desc

where Equation (2–4) is called the homogeneous system. If vector functions $\mathbf{x}_1(t)$ and $\mathbf{x}_2(t)$ are solutions of the homogeneous system, then the general solution of the varying system is

If Equations (2–4) and (2–5) are homogeneous, then they have the following form:

where $\mathbf{A}(t)$ is called the coefficient matrix, and $\mathbf{D}(t)$ the direct transference matrix.



Then we obtain

or

Figure 2-15
Mechanical system.

The output equation is

Correlation Bet

follows we shall show
system from the sta

Let us consider

This system may be

Section 2–4 / Mode

EXAMPLE 2–3

Consider again the me
are given by Equation
the state-space equati
By substituting $\mathbf{A}$

$G(s)$

[The derivation for
input vector $\mathbf{u}$ is r di
trix $\mathbf{G}(s)$ is an $m \times$

2-5 STATE-SPACE REPRESENTATION OF DIFFERENTIAL EQUATION SYSTEMS

A dynamic system can be described by ordinary differential equations. In vector-matrix notation, if a system has n order vector-matrix variables, then the system can be written as
we shall present methods for analyzing such systems.

Section 2-5 / State-Space Representation

or

where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ \cdot \\ x_n \end{bmatrix},$$

terms of the input u and its derivatives of u in the

One way to obtain the following n variable

$$x_1 = y - \beta_0$$

$$x_2 = \dot{y} - \beta_0$$

$$x_3 = \ddot{y} - \beta_0$$

.

.

.

$$x_n = y^{(n-1)} - \beta_0$$

Section 2-5 / State

$$\begin{bmatrix} x_2 \\ \cdot \\ \cdot \\ \cdot \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix}$$

y

is given also by Equ

There are many
obtaining canonical
ical form, observab
form) are presented

MATLAB can a
transfer-function rep

2-6 TRANSFORMATION OF MATHE

MATLAB is quite u
space, and vice vers
function to state spa

Section 2-6 / Trans

Another possible st

Transformation

obtain the transfer

iu must be specified

has three inputs (u

implies u_2 , and 3 in

If the system ha


```

0  0.0000  25.0000
den
1.0000  5.0000
% ***** The same result
[num,den] = ss2tf(A,B,C,D);
num =
0  0.0000  25.0000
den =
1.0000  5.0000

```

Linear Approxi

mathematical model
slightly from some c
put is $y(t)$. The rela

If the normal opera
expanded into a Tay

y

where the partial de
ating point, the high
this nonlinear syste
given by

Hence the linearized

or

When $x = 5, y = 10$,

The exact value of z is
error is 2%.

Section 2-7 / Linear

(b)

Figure 2–18

Simplified block diagrams for the system shown in Figure 2–17.

(c)

$R(s)$

Figure 2–19

Block diagram of a system.

A-2-3. Simplify the block diagram to find the transfer function $C(s)/R(s)$.

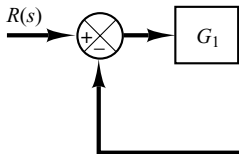
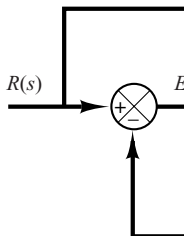


Figure 2-21
Block diagram of a
system.

Example Problems and

Figure 2–23

Control system with
reference input and
disturbance input.



R_1 —

Figure 2–24
System with two
inputs and two
outputs.

R_2 —

Example Problems and

as follows:

$$\frac{C_1(s)}{R_1(s)}$$

$$\frac{C_2(s)}{R_1(s)}$$

Note that Equations (2-25) and (2-26) contain R_1 and R_2 are present.

Notice that when $R_2 = 0$, the system reduces to those shown in Figures 2-25(a) and (b). The transfer functions are $C_1(s)/R_1(s)$ and $C_2(s)/R_1(s)$. The corresponding block diagram is shown in Figure 2-25(c).

and

where state variables

Example Problems and

Combining Equations (2-58) and (2-59) and substituting the result into Equation (2-60).

A-2-7. Obtain a state-space model for the system shown in Figure A-2-7.

Solution. From the given block diagram, we can write the following equations:

Comparing this equation with the standard form of a second-order system, we have

This is one possible solution, but there are many others. If we use MATLAB

See MATLAB Program 10.1 for a more efficient (but not equivalent.)

Example Problems and Solutions

Figure 2–26
Control system.

Figure 2-27(b).

Define the output

Figure 2-27(b) we ob

which may be modifie

Example Problems and

A-2-10. Obtain a state-space model for the system

Solution. In this problem, we have

Next, convert $K/[s(s + 1)]$ to a block diagram as shown in Figure 2-20. Then, obtain the following equations:

Solution. Referring to

In this problem, matrix

$$\mathbf{A} = \begin{bmatrix} - & \\ & \end{bmatrix}$$

Example Problems and

This system involves two transfer functions $Y_2(s)/U_1(s)$, $Y_1(s)/U_2(s)$. The first is zero and vice versa.

Solution. MATLAB
This is the MATLAB

$$z = f(\bar{z})$$

where we choose $\bar{x} =$

Since the higher-order terms, we obtain

where

$$K_1$$

$$K_2$$

$$\bar{z}$$

Example Problems and

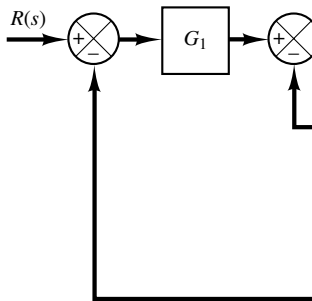


Figure 2–31
Block diagram of a system.

Figure 2–33
Control system.

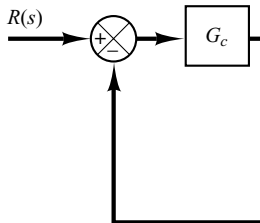


Figure 2–34
Control system.

Problems

Section 3–2 we apply the Laplace transform to differential equation models and obtain transfer functions. In Section 3–3 we obtain transfer functions for electrical, mechanical, and operational amplifiers.

3–2 MATHEMATICAL MODELING OF MECHANICAL SYSTEMS

This section first develops the basic laws for mechanical systems. Then we derive transfer functions for mechanical systems.

EXAMPLE 3-2

Let us obtain the equivalent spring constant for the system in Figures 3-2(a) and (b). The dashpot provides viscous friction between the piston and the cylinder. The dashpot provides viscous friction between the piston (or through the piston) and the cylinder. The dashpot essentially absorbs energy and does not store any kinetic energy.

By substituting E

Thus,

Hence,

Masses

Figure 3–3
Spring-mass-
dashpot system
mounted on a cart.

or

and

Equations (3–3) and (3–4) are the only state-space representations of the system.)

Section 3–2 / Math

$$\frac{X_1(s)}{U(s)} =$$

From Equations (3–6)

$$\frac{X_2(s)}{U(s)} =$$

Equations (3–7) and (

EXAMPLE 3–5

An inverted pendulum of the attitude control system is to keep the spacecraft from falling over any time.

Figure 3–5

(a) Inverted pendulum system;
(b) free-body diagram.

only a two-dimensional control force u is applied at its geometric center. C

Define the angle θ of the center of gravity c

Equations (3–16) and
They constitute a mat

EXAMPLE 3–6 Consider the inverted
centrated at the top o
case, the moment of i
 $I = 0$ in Equation (3–

Equations (3–18) and

Note that angle θ indicates the position of the cart. If we consider

(Notice that both θ and $\dot{\theta}$ are variables and Equation

Figure 3–7
Electrical circuit.

These two equations

Transfer Function

ponents that load e
is the input and e_o i

Figure 3–8
Electrical system.

The term $R_1 C_2 s$ in the numerator of two simple RC circuits is the sum of the denominators of the two simple RC circuits.

The present analysis shows that the output from the second circuit is not the same as the output from the first circuit. It is that, when we derive the transfer function, we assume that the output voltage is infinite, which means that the second circuit is connected to the first circuit. When the second circuit is withdrawn, and the transfer function of this circuit is derived, it is not valid. The degree of the numerator is not the same as the degree of the denominator. The transfer function is not valid.

which is, of course, 1

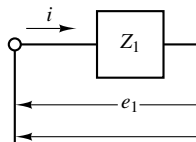
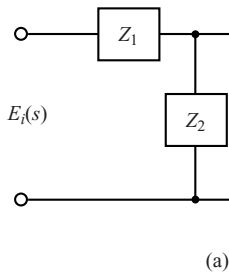


Figure 3–9
Electrical circuits.

which is the same as t

Figure 3–10
(a) The circuit of
Figure 3–8 shown in
terms of impedances;
(b) equivalent circuit
diagram.



Electronic Control
operational amplifier
amplifier circuits. The
controllers. Finally, w
the form of a table.

Figure 3–12
Electrical system.

are ideal.

Inverting Amplifier

Let us obtain the ou

Figure 3–14
Inverting amplifier.

This equation gives
circuit shown in Fig

Figure 3-15
(a) Noninverting
operational
amplifier;
(b) equivalent
circuit.

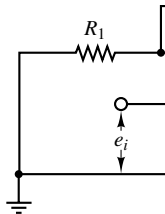
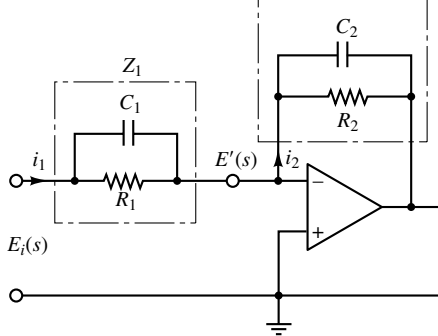


Figure 3–16
First-order lag circuit
using operational
amplifier.

The transfer function

which is, of course, the

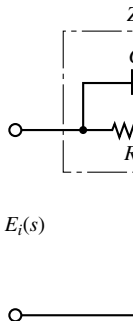


(a)

Figure 3–18

(a) Operational-amplifier circuit; (b) operat

Figure 3–19
Electronic PID
controller.



K_p is called the proportional gain. K_d is called the derivative gain. For

Table 3–1 shows the transfer functions for the various controllers or compensators.

6	Lead or lag	$\frac{R_4}{R_3} \frac{R_2}{R_1} \frac{R_1 C_1 s}{R_2 C_2 s}$
7	Lag-lead	$\frac{R_6}{R_5} \frac{R_4}{R_3} \frac{[(R_1 + R_3) C_1 s + 1]}{(R_1 C_1 s + 1) [(R_2$

Figure 3–20
(a) Automobile
suspension system;
(b) simplified
suspension system.

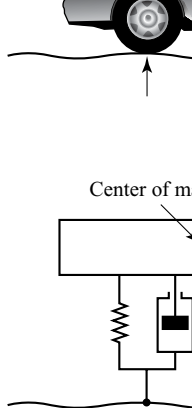


Figure 3–21
Suspension system.

Example Problems and

and the output equation

- A-3-4.** Obtain the transfer function for the system shown in Figure A-3-4. Obtain the transfer function for the system shown in Figure A-3-4. Obtain the transfer function for the system shown in Figure A-3-4.

Hence the transfer function

For the electrical system

Example Problems and

Figure 3–25

(a) Bridged T network in terms of complex impedances;
(b) equivalent network.

Similarly, for the bridge

into Equation (3–38).

Example Problems and

Hence

A-3-7. Obtain the transfer function $E_o(s)/E_i(s)$ of the op-amp circuit in Fig. P-3-7, using complex impedances.

Solution. From Fig. P-3-7, we have

To find the transfer fu

into Equation (3–41).

which is, as a matter o

Example Problems and

$$\frac{E_o(s)}{E_i(s)} =$$

 e_i

Figure 3–28
Operational-
amplifier circuit.

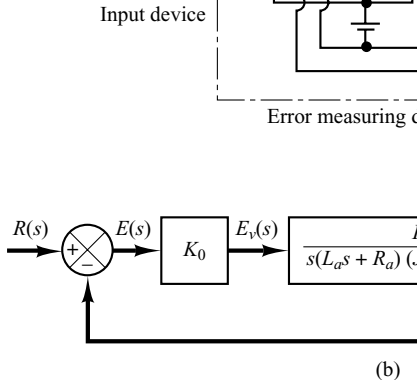


Figure 3-29

(a) Schematic diagram of servo system;

Example Problems and

$$G(s)$$

When L_a is small, it c
becomes

The term $[b_0 + (K_2K$
viscous friction of the



Figure 3–30
Damper system.

Problems

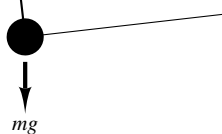


Figure 3–33 Spring-loaded pendulum system

B–3–5. Referring to Examples 3–5 and 3–6, consider the inverted-pendulum system shown in Figure 3–34. Assume that the mass of the inverted pendulum is m and is distributed along the length of the rod. (The center of gravity of the pendulum is located at the center of the rod.) Assuming that θ is small, derive mathematical models for the system in the forms of differential equations, transfer functions, and state-space equations.

B-3-11. Obtain the transfer function $E_o(s)/E_i(s)$ for the op-amp circuit shown in Figure 3-40.



e_i



Figure

Problems

tomation of product
pneumatic circuits th
wide usage. Also, var
we present hydraulic
control systems, etc
controllers. Both pne
the concepts of resis
systems involve heat
such systems can be

Outline of the
chapter. Section 4
systems—in particu
discusses hydraulic
Section 4–5 analyze

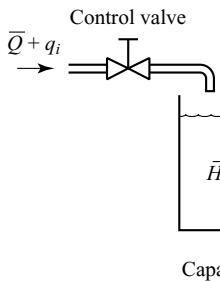


Figure 4-1
 (a) Liquid-level
 system; (b) head-
 versus-flow-rate
 curve.

By use of the tu
given by

Such linearization i
respective steady-st

In many practical
on the flow coefficient
be determined by p
and measuring the sl
is shown in Figure 4-
gent line to the curv
of this tangent line i
 $2\bar{H}/\bar{Q}$, the resistance

From the definition

The differential equation

Note that RC is the time constant. Multiplying both sides of Equation (4)

where

Section 4–2 / Liquid Level Control

$$\bar{Q} + q \rightarrow \Rightarrow$$

Figure 4-2
Liquid-level system
with interaction.

Figure 4–3

(a) Elements of the block diagram of the system shown in Figure 4–2; (b) block diagram of the system; (c)–(e) successive reductions of the block diagram.

are always needed.

8. Normal operation

The pneumatic range. Pneumatic hydraulic systems operate. Normal temperature. Normal to 158°F).

9. Pneumatic systems

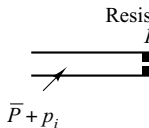
not, unless not

In what follows we shall present pneumatic systems.

We shall first give controllers operate control actions. The

Figure 4-4

(a) Schematic diagram of a pressure system;
(b) pressure-difference-versus-flow-rate curve.



n is equal to the ratio of the pressure and c_v is the specific heat. n is approximately equal to 1.4.

The value of γ is 1.4. Substituting Equation (4–10) into Equation (4–9) yields

or

Substituting Equation (4–11) into Equation (4–9) yields

or

which can be written

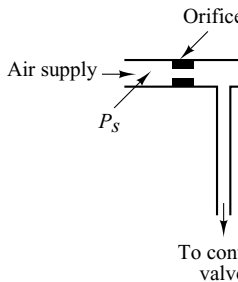
If p_i and p_o are constants
of the system is

where RC has the dimension

Section 4–3 / Pneumatics

Figure 4-5

(a) Schematic diagram of a pneumatic nozzle-flapper amplifier;
(b) characteristic curve relating nozzle back pressure and nozzle-flapper distance.



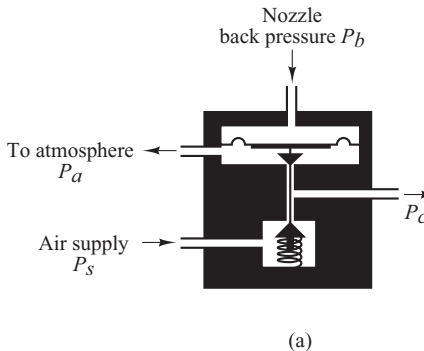


Figure 4-6

(a) Schematic diagram of a bleed-type re

be regulated by means of a flapper connecting both the error signal

The operation of the control signal to the two-stage actuating error signal, nozzle back pressure, increase of the control the flapper to the right flapper displacement.

It should be noted that back bellows move to the (If these two movements

Equations for the error zero, or $e = 0$, an equation

Figure 4–8

- (a) Schematic diagram of a force-distance type
 (b) flapper mounted on a fixed point; (c) flapper
 (d) displacement x as a result of addition of two
 (e) block diagram for the controller; (f) simplified

displacement of bellows
 nozzle back pressure
 error exists, the nozzle
 of the diaphragm, the
 respective equilibrium
 itive direction for each

e is given by

A simplified block diagram of the pneumatic controller is shown in Figure 4-10. As seen from Equation (4-10), the gain of the controller is widely varied by adjusting the feedback bellows pressure. The gain setting range is not shown in Figure 4-10, but is usually obtained by a gain adjusting knob or other means. As noted earlier, the gain of the controller is the feedback bellows pressure divided by the setpoint pressure.

pressure P_o are fed
 matic controller op
 the reference input

Figure 4–10
 Schematic diagram
 of a force-balance
 type pneumatic
 proportional
 controller.

Atmos
 Refer
 input pr
 C
 pr
 Air s

we may neglect the
reflect this assumpt

and the transfer fun

where p_e is defined
proportional contro
the value of k depe
the feedback chamb
orifice of the inlet p

Figure 4–11
Schematic diagram
of a pneumatic
actuating valve.

$$\|G(s)H(s)\| \gg 1,$$

Thus, if proportion
having the transfer

Figure 4–12
Control system.

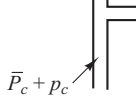


Figure 4-14

(a) Pneumatic proportional-plus-derivative controller;
 (b) step change in e and the corresponding changes in x and p_c plotted versus t ; (c) block diagram of the controller.

$E(s)$

the proportional co-
variables, we can sh
proportional contro

Consider the pneumatic controller is as follows: "without any restriction, the source through a restriction will cause the back pressure control pressure p_c to follow the path to bellows II, the flow across the valve is p_c . Thus bellows II will flapper an additional amount the back pressure p_c ."

diagram of the
controller;
(d) simplified block
diagram.

Note that the $1/s$ term in the denominator cancels the feedback gain K .

A block diagram of the closed-loop system with the same variables is shown in Figure 4–15(d). The transfer function is

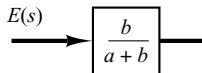


Figure 4-16

(a) Pneumatic proportional-plus-integral-plus-derivative controller;
(b) block diagram of the controller.

load forces. For such

Hydraulic Systems

applications, aircraft
tors as positiveness
stopping, and revers

The operating pr
(between 1 and 35 M
to 10,000 lb_f/in.² (7
the hydraulic unit c
pressure hydraulic
positioning of heav
tronic and hydraulic
electronic control a

fluid can greatly affect the results. Therefore, the tests must be carried out with care.

Note that most of the models used to linearize nonlinear systems are sufficiently accurate only for small variations with nonlinear systems.

Hydraulic Servo

essentially a pilot-valve system. The valve is a balanced valve. A very large power is required to move the valve with very little power.

In practice, the pilot valve is used to move the responding valves. In

age improves both the accuracy and the resolution of the following analysis with respect to the flow rates through the valves—that is, the flow rates through the high-frequency signal valves. The high-frequency signal improves the sensitivity of the analysis to the displacement of the valves.

We shall apply the linearized mathematical model to the underlapped and symmetrical valves of the power cylinder that is used to move a load.

In Figure 4–17(1) we define the valve orifice area A_v and the flow rates through the valves.

Section 4–4 / Hydraulics

this q and is given by

$$q = q$$

In the present case, since the metric is symmetrical, we have

or

If we define the pre-

Coefficients a and b are determined by the mathematical model of the valve. The values of valve a and b are determined by the valve. a is negative and so b is positive.

Since the normal operating pressure is p_0 , the normal operating pressure is p_0 .

where

x

$x =$

Figure 4–18
Characteristic curves
of the linearized
hydraulic
servomotor.

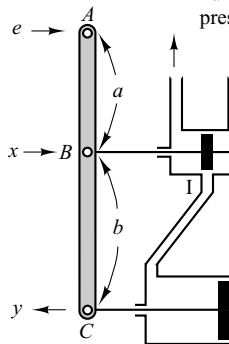
It is noted that a β (including the effect of β into consideration,

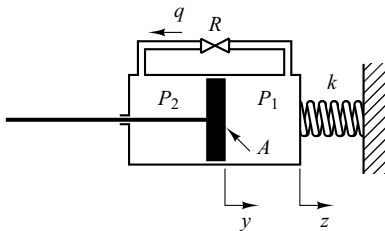
where T_1 and T_2 are the volume of oil in constants.

Section 4–4 / Hydr

Figure 4-19
Hydraulic
servomotor.

Figure 4-20
 (a) Servomotor that acts as a proportional controller; (b) block diagram of the servomotor.

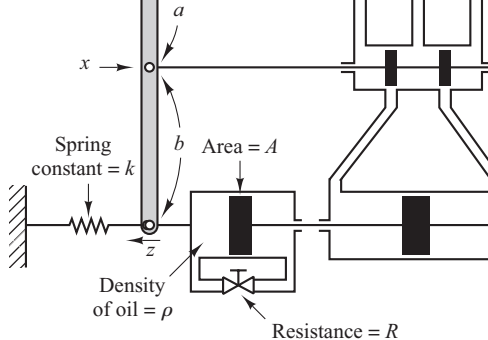




(a)

Figure 4–21

(a) Dashpot; (b) step change in y and the corresponding diagram of the dashpot.

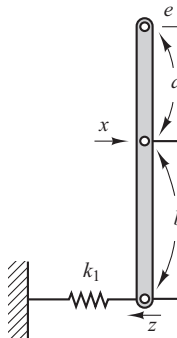


(a)

Figure 4–22

(a) Schematic diagram of a hydraulic proportional

Figure 4-24
Schematic diagram
of a hydraulic
proportional-plus-
integral-plus-
derivative controller.



4-5 THERMAL SYSTEMS

Thermal systems a
another. Thermal s
although the therm
accurately as lumpe
stance. For precise a
to simplify the anal
lumped-parameter
flow have negligible
capacitance have ne

Since the thermal capacity of a body is the product of its mass and specific heat, the thermal resistance for a body of mass m and specific heat c is

The thermal capacity of a body is

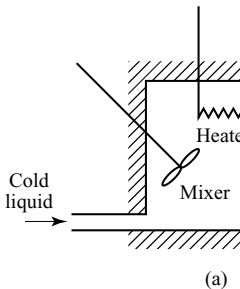
or

where m = mass of the body
 c = specific heat of the body

Section 4-5 / Thermal Resistance

Figure 4-26

(a) Thermal system:
(b) block diagram of
the system.



where $\Theta(s) = \mathcal{L}[\theta]$

If the present temperature of the inflowing liquid and the temperature change θ in the temperature of the inflowing liquid, the equation:

A block diagram of a control system involves two blocks:

Section 4–5 / Thermal Systems

or

Thus, the head becomes

9

Figure 4–27
Liquid-level system.

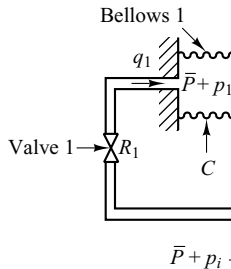
$$\bar{Q}_1 + q_{i1} \longrightarrow$$

Figure 4–28
Liquid-level system.

Example Problems and

expand or contract linearly with the constant of the bellows system.

Figure 4–29
 (a) Pneumatic pressure system;
 (b) pressure-difference-versus-mass-flow-rate curves.



The numerical value of

$$C =$$

where $R_{\text{air}} = 287 \text{ N}\cdot\text{m}$

By substituting the nu

Example Problems and

Figure 4–30
Schematic diagram
of a pneumatic
controller.



$E(s)$



Figure 4–31

(a) Block diagram of the pneumatic controller shown in Figure 4–30;
(b) simplified block diagram.

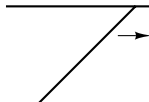
Example Problems and

Solution. Define the displacement of the p

A

Figure 4-33

(a) Uncovered-port-area- A -versus-displacement- x curve for the overlapped valve; (b) uncovered-port-area- A -versus-displacement- x curve for the underlapped valve.



(

where A is the power

where $K = K_1/(A\rho)$

The controller produc

Example Problems and

the transfer function

The speed controller

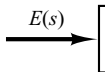


Figure 4–36
Block diagram for
the speed control
system shown in
Figure 4–35.

Substituting Equation

Taking the Laplace tr

Figure 4-37
Hydraulic system.



Example Problems and

Let us assume that the
temperature is suddenly
changed from $\bar{\theta}_o$ to θ_o .
The equation describing

Figure 4–38
Air heating system.

rows. The heat enters the thermometer. This heating changes its temperature by

Figure 4–40
Thin, glass-wall,
mercury thermo-
meter system.

Example Problems and

Figure 4-42. The flow through the valve is turbulent and is related to the head H by

$$Q = 0.005\sqrt{H}$$

where Q is the flow rate measured in m^3/sec and H is in meters.

Suppose that the head is 2 m at $t = 0$. What is the head at $t = 60$ sec?

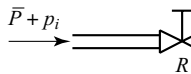


Figure 4-44
Pneumatic system

Problems

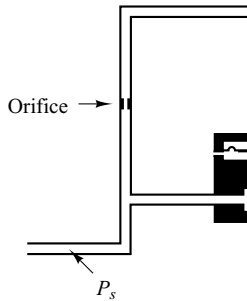


Figure 4-46
Pneumatic controller.

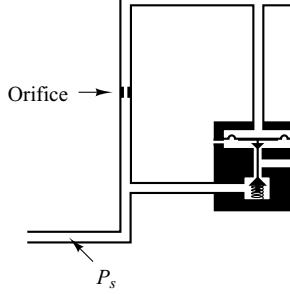


Figure 4-48
Pneumatic controller.

Problems

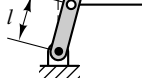


Figure 4–51
Aircraft elevator
control system.

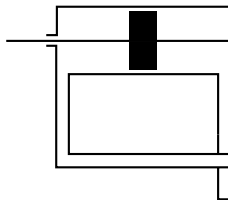


Figure 4-52

Liquid-level control s

Problems

In analyzing and performance of various test input signals and

Many design criteria response of systems to test signals can be just characteristics of a system with actual input signals

Typical Test Signals
ramp functions, acceleration noise. In this chapter signals. With these tests systems can be carried

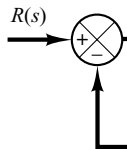
Important systems require careful consideration. A control system involving a time-varying input, cannot follow a constant input. The steady state can be reached only if the system exhibits damped oscillations. If the steady state does not exist, there is a steady state error. This error can be reduced by a feedback system, we must examine the system.

Outline of the chapter
The chapter deals with aperiodic signals (step, ramp, and parabolic) and their response. The outline of the chapter is as follows: Section 5-1 deals with the step response of a second-order system. Section 5-2 deals with the ramp response of a second-order system. Section 5-3 deals with the parabolic response of a second-order system.

Equation (5-3) states
 One important characteristic
 the value of $c(t)$ is 0
 may be easily seen by

Figure 5-1

(a) Block diagram of
 a first-order system;
 (b) simplified block
 diagram.



the unit-ramp func

Expanding $C(s)$ into

Taking the inverse

The error signal $e(t)$

Figure 5-4
Unit-impulse
response of the
system shown in
Figure 5-1(a).

conditions, we obtain

So the transfer function

By using this transfer function, the system can be modified to the form obtained as

$$\frac{C}{R}$$

Such a system when compared with a second-order system

$$\frac{C(s)}{R(s)} = \frac{K}{s^2 + Bs + K}$$

The closed-loop po
 $B^2 - 4JK \geq 0$. In t

where σ is called th
ing ratio of the syste
critical damping B_c

Section 5-3 / Seco

ten in the following

C

Referring to the La

(2) *Critically damped*
is said to be a critically
For a unit-step i

The inverse Laplace

This result can also
using the following

Section 5–3 / Seco

This approximate final values of both other.

With the approximation obtained as

The time response of

This gives an approximation be neglected.

Definitions of

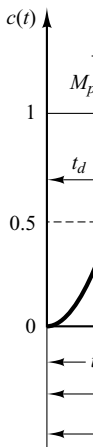
ance characteristics
a unit-step input, since
a step input is known

The transient re
tions. For convenien
mon practice to use
the output and all t
many systems can b

The transient re
tions before reachin
a control system to

1. Delay time, t_d
2. Rise time, t_r

Figure 5–8
Unit-step response
curve showing t_d , t_r ,
 t_p , M_p , and t_s .



Thus, the rise time t_r

where angle β is de

Figure 5–9
Definition of the
angle β .

Section 5–3 / Seco

The maximum percent

If the final value
equation:

Settling time t_s : For
obtained from Equa

$$c(t) = 1$$

the 5% criterion)
The discontinuities
in the value of ζ can

For convenience
settling time t_s to be

or

Note that the se
ratio and the undan
determined from th

Section 5–3 / Seco

M_p

Figure 5–12
 M_p versus ζ curve.

For the 5% criterion,

Servo System

be used to improve position signal, it is output signal. (No discontinuous noise than the useful signal voltage signal because are induced in the potentiometer therefore

Section 5-3 / Seco

The undamped natural frequency is $\omega_n = 1$ rad/sec, indicating that the maximum overshoot is the value of the damping ratio. For the velocity-feedback system, the damping ratio is $\zeta = 1$.

It is important to note that the damping ratio without feedback is $\zeta = 0.5$.

EXAMPLE 5-2

For the system shown in Fig. 5-10, determine the constant K_h so that the system has a damping ratio of 0.5. With these values of K and K_h , determine $B = 1$ N-m/rad/sec.

Determination of the value of K_h from (5-21) as

where

Thus, t_r is

Settling time t_s : For $t > t_s$

For the 5% criterion,

$$\frac{c(t)}{\omega_n}$$

Figure 5–14
Unit-impulse
response curves of
the system shown in
Figure 5–6.

response function, found from the correlation impulse response coefficient, 5–15, is $1 + M_p$, where M_p is given by Equation (5–19). Equation (5–20) can be used to find the time axis.

5–4 HIGHER-ORDER SYSTEMS

In this section we shall discuss the general terms. It will be shown that the responses of first-order

Section 5–4 / Higher-Order Systems

where a_i is the residue at s_i . If $C(s)$ is then $C(s)$ will have the form given by Equation 5-10. The command `residuez` can be used to find the residues. (See Appendix B.)

If all closed-loop poles are real, the residues determine the system response.

Figure 5–16
Control system.

the damped exponential. The steady-state output is $C(0)/R(0)$. Let us assume that the poles $p_1, p_2, \dots, p_n$ are located far from the origin, so that the terms that correspond to them are negligible for a horizontal distance from the origin. The steady-state output is due to that pole at the origin.

Remember that the poles $p_1, p_2, \dots, p_n$ are the poles, while the zeros $z_1, z_2, \dots, z_m$ are the loop zeros. As we have seen, the steady-state response terms in the transient-response time constant of $C(s)/R(s)$ do not depend on the magnitudes and signs of the poles and zeros.

Section 5-4 / Higher-Order Systems

including the $j\omega$ axis oscillations, the amplitude cases, where noise is rate determined by closed-loop poles of

Note that the model guarantee satisfactory closed-loop poles lie oscillations or may be response characteristic particular region in in Figure 5–17.

Since the relative control system are direct

is represented by two

This system can be
polynomials in decr

An alternative repr

Section 5-5 / Trans

no plot is shown on the response curves. The system, respectively, even outputs and one row for each element in

Note in Equation and specifies which If the system involves given by Equation and output combination

(For details, see Exa

Hence

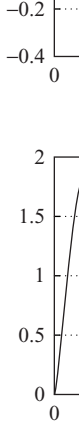
Since this system is
depending on which s

Section 5–5 / Trans

Figure 5–18
Unit-step response
curves.

Amplitude

To: Y2



```

step(A,B,C)
grid
title ('Step
text(3.4, -0.
text(3.4, 1.

% ***** N
% is u2. E

step(A,B,C)
grid
title ('Step
text(3,0.14
text(2.8,1.

```

Unit step response curves. (a) u_1 is the input ($u_2 = 0$); (b) u_2 is the input ($u_1 = 0$).

Writing Text on
for example, the fol

and

The first statement t
 $y = -0.06$. Similarly
the coordinates $x =$

Obtaining the $\mathbf{C}$
consider the unit-stress

Section 5-5 / Trans

Figure 5–20
Unit-step response
curve.

Notice in Figure 5–20 that the plot is automatically determined by the `step` function. To modify the step command, use the `options` command and the `y` axis as ‘Output’ or ‘Response’, such as

`options y='Output';`
or, more generally,

`options y='Output';`
and use `plot(t,y)` command.

MATLAB. MATLAB uses the `mesh` command to obtain three-dimensional “mesh” plots and the `surf` command to fill in the spaces between the “mesh” command.

EXAMPLE 5–4 Consider the closed-loop transfer function

(The undamped natural frequency $\omega_n = 1$ rad/sec and the damping ratio ζ assumes the following values.)

Also plot a three-dimensional surface plot of the magnitude of the transfer function.

Section 5–5 / Transient Response of Second-Order Systems

```

ylabel('Response')
text(4.1,1.86,'\zeta = 0')
text(3.5,1.5,'0.2')
text(3.5,1.24,'0.4')
text(3.5,1.08,'0.6')
text(3.5,0.95,'0.8')
text(3.5,0.86,'1.0')

% To plot a three-dimensional
mesh(t,zeta,y')
title('Three-Dimensional Plot of')
xlabel('t Sec')
ylabel('\zeta')
zlabel('Response')

```

response curves for $\zeta = 0, 0.2, 0.4, 0.6, 0.8$, and 1.0; (b) three-dimensional plot of unit-step response curves.

**Obtaining Rise
with MATLAB.** M
maximum overshoot

MATLAB Program
time. A unit-step re

Section 5–5 / Trans

Figure 5–23
Unit-step response
curve.

If MATLAB is
[y,x,t] = impulse(A,
system and the time
tain the output and
as many columns as
as state variables a
must include a plot

EXAMPLE 5–5 Obtain the unit-impulse

response of $sG(s)$.
 Consider the un
 $R(s) = 1$ for the un

We can thus conve
 $sG(s)$.
 If we enter the f

divide $G(s)$ by s and
loop system

For a unit-ramp input

To obtain the unit-ramp
response, divide the
numerator into the

Section 5-5 / Trans

Figure 5–26
Unit-ramp response
curve.

where

$$\mathbf{AA} =$$

$$\mathbf{BB} =$$

Note that x_3 is the t
be obtained by ente
ramp response curv

Section 5–5 / Trans

Figure 5–27
Unit-ramp response
curve.

Figure 5–28
Unit-ramp response.


```

xlabel('t Sec')
ylabel('Output')

% For the response
% "z = lsim(A,B,C,u,t);
u = exp(-t);
z = lsim(A,B,C,u,t);
plot(t,u,'-',t,z,'-');
grid
title('Response')
xlabel('t Sec')
ylabel('Exponential')
text(2.3,0.49,'1/e')
text(6.4,0.28,'1/e^2')

```

Figure 5–29

- (a) Unit-step
response;
(b) response to input
 $u = e^{-t}$.

–0

Response to Input

for obtaining the response to a given input or “initial”. We shall now obtain the response to a given input using a simple method. We shall first obtain the response to a given input when the system is given a unit-step input. We shall then obtain the response to a given input when the system is given a unit-step input.

Section 5–5 / Transfer Function

```

% ***** Sy
% a unit-st

% ***** En
% function

num = [0.1
den = [1 3

% ***** En

step(num,d

% ***** En

grid
title('Respo

```

(Notice that by taking the inverse Laplace transform of a differential equation

Now define

Then Equation (5–5)

By integrating Equation

where

Section 5–5 / Trans

(See, for example,

Case A. When the
terminated automatic

Section 5-5 / Trans

where

$$\mathbf{A} = \begin{bmatrix} - \\ - \end{bmatrix}$$

$$\mathbf{x}_0 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Section 5-5 / Trans

Figure 5–33

Response curves to
initial condition.

EXAMPLE 5–10

Consider the following
function is present.)

Obtain the response y

Output y

1

0

Figure 5–34
Response $y(t)$ to
initial condition.

all the coefficients
 from the follo
 ways be fact
 $(s^2 + bs + c)$
 the quadratic
 $(s^2 + bs + c)$
 itive. For all ro
 factors must b
 containing on
 coefficients. It
 positive is no
 condition for s
 all have a pos
 multiplying bo

pattern of cross-mu
evaluating the c 's, d

Section 5–6 / Rout

The condition that all

EXAMPLE 5-12 Consider the following

Let us follow the pro
two rows can be ob

indicates that there

the array of coefficient

There are two sign c
in the right-half s p
form of the polynom

Section 5–6 / Rout

s^3 s^2 s^1 s^0

We see that there is
inal equation has on
polynomial equation

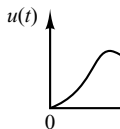
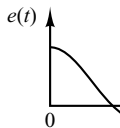
we obtain

or

Figure 5–35
Control system.

Figure 5-36

(a) Plots of $e(t)$ and $u(t)$ curves showing nonzero control signal when the actuating error signal is zero (integral control); (b) plots of $e(t)$ and $u(t)$ curves showing zero control signal when the actuating error signal is zero (proportional control).



Such a system with no error in the step response has no offset in the unit-step response.

Figure 5–38
Unit-step response
and offset.

Assuming that the transfer function $C(s)$ and $D(s)$ is given by

R

Figure 5–40

Control system with
a torque disturbance.

In the absence of the

Figure 5-41

Proportional-plus-integral control of a load element consisting of moment of inertia and viscous friction.

$R = 0$
→

Section 5-7 / Effect

will have roots with negative real parts, and the system will be stable in practice.

Note that in the proportional control case, to stabilize the system, the gain must be large enough to make the state error in response to a step change small.

Derivative Control Derivative control is a proportional control with an added sensitivity to the rate of change of the error. An advantage of derivative control is that it anticipates the action required to stabilize the system and thus increases the stability margin.

are imaginary, the root locus is shown in Figure 5–4.

Control systems
see that the addition

Proportional-P
modify the proportion
transfer function is
to $K_p(e + T_d\dot{e})$. Der
error velocity, and
appropriate counter

Section 5–7 / Effect

The characteristic e

$R(s)$
→

Figure 5–45
Control system.

function $G(s)$:

It involves the term s^N in the denominator of $G(s)$ of the origin. The present case is $N = 1$, $N = 2, \dots, \infty$ by the open-loop transfer function. $N = 1, N = 2, \dots, \infty$ order of a system. $N = 1, N = 2, \dots, \infty$ increasing the type number. $N = 1, N = 2, \dots, \infty$ steady-state accuracy.

We shall see later that the numerator of $G(s)$ is the denominator, except for a constant factor. The loop gain K is directly proportional to the steady-state accuracy.

Section 5-8 / Steady-State Accuracy

unit-step input is

The static position of

Thus, the steady-state

The static velocity e

Thus, the steady-sta

The term *velocity*
input. The dimension
error is not an error
For a type 0 system

Section 5–8 / Stead

The foregoing analysis shows that the system response to a unit-step input in the steady state is a constant value, $1/K$, with a finite error. In the transient response, the system follows the input velocity, but the error is not zero. The error is the input and is inverse to the system response of a type 1 system can follow a unit-step input.

Static Acceleration Error
with a unit-parabolic input.

For a type 3 or high

$$K_a =$$

Thus, the steady-sta

Remember that steady-state deviations from the desired value for transients have died out. The error constant is a finite position difference.

The error constant is used to design a system to reduce or eliminate the steady-state error and improve performance. It is good to know how to design the transient response and the steady-state performance with integrators to the feedforward path. The design of the feedforward path is a problem. The design of the feedforward path is a problem.

Figure 5–49
Mechanical system.

Example Problems and

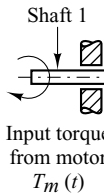


Figure 5–51
Gear-train system.

Thus, the equivalent J to shaft 1 are given, re

Similarly, the equivalent J to the load shaft (shaft

Example Problems and

Figure 5–52

(a) Closed-loop system; (b) unit-step response curve.

from which

$$\xrightarrow{R(s)}$$

Figure 5–53
Closed-loop system.

Example Problems and

Figure 5-54
 (a) Mechanical
 vibratory system;
 (b) step-response
 curve.



Thus, the logarithmic

It is a function only of
of the logarithmic. de

In the experiment
lation, we measure the
it is necessary to choo

Example Problems and

It follows that at $t = 1$

Consequently, the am

A-5-8. Obtain both analytical
system:

[Obtain the partial-fra

-0.2813+ 0.1711i
 -0.4375
 -0.3750
 1.0000

p =

-2.0000+ 4.0000i
 -2.0000- 4.0000i
 -2.0000
 -2.0000
 0

k =

[]

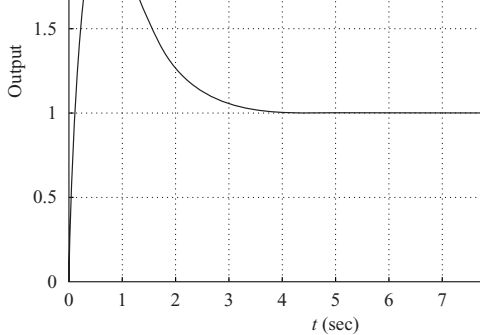
Example Problems and

A-5-9. When the closed-loop system may exhibit a large overshoot, use MATLAB:

Obtain also the unit-ramp response.

Solution. MATLAB plots the response of the system to the unit-ramp input, as shown in Figure 5-35.

Notice that the unit-ramp response curve leads the derivative term in the error signal.



(a)

Figure 5-57

(a) Unit-step response curve; (b) unit-ramp response curve

Example Problems and


```
peak_time =  
peak_time =  
1.6600  
max_oversho  
max_oversho  
0.6182  
s = 1001; wh  
settling_time  
settling_time  
10.0200
```

Case 2: $\zeta =$

Case 3: $\zeta =$

Case 4: $\zeta =$

Solution. Define $\omega_n^2 =$

Example Problems and

Figure 5–59
Unit-step response
curves for four cases.

```
% --
```

```
num
```

```
den
```

```
t = 0
```

```
r1 =
```

```
y1 =
```

```
plot(
```

```
grid
```

```
title(
```

```
xlab
```

```
ylab
```

```
text(
```

```
text(
```

Example Problems and

response curve,
(b) response to
exponential input
 $r_1 = e^{-0.5t}$.

A-5-13. Obtain the response of

when the input $r(t)$ is

[The input $r(t)$ is a step

$$r(t) = 2 + t.$$

A-5-14. Obtain the response of

[The input $r(t)$ is the

Figure 5-62
Control system.

Example Problems and

Figure 5–63

Response to unit-acceleration input.

A–5–15. Consider the system of

```
% We shall show two three-dimensional plots  
% "mesh(y)". These two plots are  
% interchanged.
```

```
mesh(y)  
title('Three-Dimensional Plot of U')  
xlabel('n, where n = 1,2,3,4,5,6')  
ylabel('Computation Time Points')  
zlabel('Outputs')
```

```
mesh(y')  
title('Three-Dimensional Plot of U')  
xlabel('Computation Time Points')  
ylabel('n, where n = 1,2,3,4,5,6')  
zlabel('Outputs')
```

Example Problems and

(b)

A-5-16. Consider the system s

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} =$$

$$y =$$

(There is no input or
given initial condition

the given initial
condition.

A-5-17. Consider the following

Determine the range

Solution. The Routh

Example Problems and

where we substituted
 sary and sufficient th
 minors are the follow

where $a_s = 0$ if $s > n$
 are included in the co
 positive, and $a_0 > 0$ a

Solution. The Routh

Example Problems and

then

In any case, the last te

where $a_1, b_1, c_1, \dots$, and

Consider, for exam

The determinant is un
from the second row a

Example Problems and

Finally, subtracting from

where

Hurwitz conditions for

Thus we have demon
Routh's conditions fo
determinants of any o
ity criterion can be es

A-5-21. Consider the character

Using the Hurwitz sta

Solution. Comparing

Example Problems and

Solution. Consider, for $t > 0$, the response of the system to a nonzero constant input r . The steady-state error is the assumption that $c = r$ in the steady state.

A nonzero offset r in the input results in a nonzero steady state, if e were zero. This results in the assumed steady state of the system.

Figure 5–66
Control system.

From this analysis
member of the system
cel the disturbance to
speed so that nonzero

Figure 5–68

Block diagram of the
speed control system
of Figure 5–67 when
 $\Omega_r(s) = 0$.

Example Problems and

presents a stability plot.
One method of st
choose

$\Omega_r(s)$

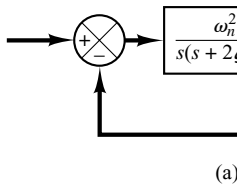
Figure 5–69

Block diagram of a
speed control system.

Then

Figure 5–71

(a) Control system;
(b) control system
with input filter.



Example Problems and

and

Then

and

For a unit-step input,

B-5-1. A thermometer requires 1 min to indicate the response to a step input. Assuming the thermometer is a first-order system, find the time constant.

If the thermometer is placed in a bath, the temperature of which is changing linearly at a rate of $10^\circ/\text{min}$, what error does the thermometer show?

B-5-2. Consider the unit-step response of a unity feedback control system whose open-loop transfer function is

$$G(s) = \frac{1}{s(s + 1)}$$

Problems

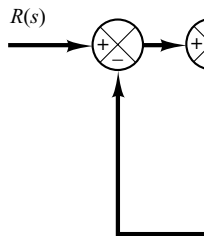


Figure 5-74
(a) Control system;

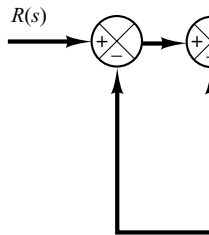


Figure 5-76
Block diagram of a

Problems

Figure 5-77

Positional servo system (System I), action (System II), and position (System III).

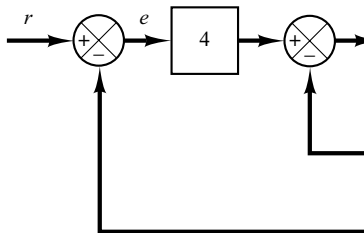


Figure 5-78

Position control system.

Figure 5–80

(a) Unstable satellite attitude control system
(b) stabilized system.

Problems

Figure 5-81

Servo system with tachon

From the design of the closed-loop poles to the selection of an appropriate result, addition of a discussed in detail in

The closed-loop characteristic equation is the solution. (MATLAB) the roots of the characteristic equation are the open-loop transfer function. Computations must

A simple method was developed by W. R. Rife, called the *root-locus* method.

Angle and Mag
in Figure 6–1. The c

Figure 6–1
Control system.

Then the root loci for $G(s)$ varied from zero to infinity.

Note that to begin with, we must know the location of the poles and zeros of the complex quantity $G(s)$. The test points s are not on the root loci. The test point s is given by

of the root loci and

Illustrative Exa

root-locus plots will
of the root loci are c
with inspection, to
equation of the clo
understanding of h
loop poles and zeros
purposes, the proce
order systems.

Because graphic
sis, we find it necess
when sketching the

and the angle condition
 -1 forms a portion of

and

Figure 6–3
Control system.

Section 6–2 / Root-

A root-locus diagram
seen as follows: The e

or

which can be written a

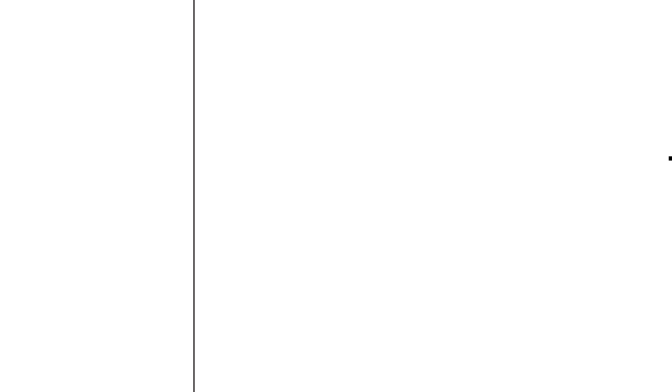


Figure 6-4
Three asymptotes.

and

If dK/ds is set equal to zero, the breakaway points can be similarly determined.

It should be noted that the breakaway point is not necessarily the actual breakaway point. It may be a break-in point or a breakaway point. If dK/ds is a positive value, then the system is stable.

which yields

The frequencies at the poles are those corresponding to the characteristic frequencies of the system.

An alternative approach is to consider the real part and the imaginary part of the characteristic equation, which yields

or

Equating both the real and imaginary parts to zero yields

Section 6-2 / Root Locus

The figure area is mostly blank, suggesting the root-locus plot was not rendered or is obscured. A vertical line is present on the left side of the page, separating the caption from the page number.

Figure 6–6
Root-locus plot.

1. *Determine the root locus angular contributions*
 The effect of the complex-conjugate poles on the real axis is determined by a section of the negative real axis that, since this locus has two root loci, each of which has a root loci break in the

Figure 6–7
 Control system.

Figure 6–9
Determination of the
angle of departure.

A root-locus plot in the complex s-plane. The horizontal axis is the real part (σ) and the vertical axis is the imaginary part (jω). There are two poles marked with 'x' on the real axis, one at the origin (0,0) and one at a negative value. There is one zero marked with a 'o' on the real axis at a positive value. The root locus branches start at the poles and end at the zero. One branch is on the real axis between the origin and the zero. Another branch is on the real axis to the left of the pole at the negative value. A third branch starts at the pole at the origin and moves into the complex plane, curving to the left and upwards. A fourth branch starts at the pole at the negative value and moves into the complex plane, curving to the left and downwards. The two complex branches are symmetric about the real axis and appear to meet at a breakaway point on the real axis between the origin and the zero, then curve away from each other into the complex plane.

Figure 6–10
Root-locus plot.

or

which can be simplified

or

This last equation is e

cable to systems with

1. *Locate the poles from open-loop polynomial form of the transfer function in the s plane. [Note: closed-loop zeros cancel out]*

Figure 6–11
Control system.

where n = number

m = number

Here, $k = 0$ corresponds to the asymptote at $\omega = 0$, though k assumes a value of $k = 1, 2, \dots$ and the number of corner frequencies is k .

All the asymptotic approximations so is obtained as follows. The corner frequencies of the transfer function are expressed in rad/sec as

$$G(s)H(s)$$

The breakaway point is determined by the characteristic equation. Hence, the breakaway point can be determined by

where the prime indicates differentiation with respect to ω . The breakaway point is the value of ω at which the root locus has all roots of Equation (6-14) lying on the real axis. The breakaway point is the break-in point. If a root locus has a breakaway point on the real axis, then this root locus is a breakaway point.

and the origin. The
the asymptotes but i

Figure 6–12

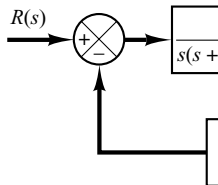
Construction of the
root locus. [Angle of
departure
 $= 180^\circ -$
 $(\theta_1 + \theta_2) + \phi.$]

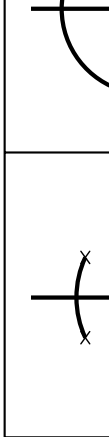
slight change in the
look quite different

Figure 6–13
Root-locus plots.

Figure 6–14

(a) Control system with velocity feedback; (b) and (c) modified block diagrams.





Section 6-2 / Root-

Note that both vectors $\mathbf{p}$ and $\mathbf{z}$ are column vectors.
A MATLAB command to compute the closed-loop

Using this command, the closed-loop poles are automatically determined. The closed-loop system loop poles are to be determined.

For the systems $\mathbf{p}$ and $\mathbf{z}$ are column vectors.
system with the gain K is determined.
Note that command `roots` can be used to

use the user-supplied `roots` function.

The denominator

Figure 6–15
Control system.

The figure area is mostly blank, suggesting the root-locus plot was not rendered or is obscured. A vertical line is present on the left side of the page.

Figure 6–16
Root-locus plot.

Figure 6–17
Root-locus plot.

```

K = [K1 K2]
r = rlocf(K, p)
plot(r, 'o')
v = [-4 4]
grid
title('Root Locus')
xlabel('s')
ylabel('j\omega')

```

EXAMPLE 6–5 Consider the system shown in Figure 6–10.

MAT

% --

A =

B =

C =

D =

K =

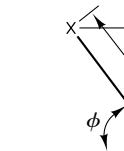
rloc

v =

grid

title

Section 6–3 / Plotti



x

Figure 6–21

(a) Complex poles;
(b) lines of constant
damping ratio ζ .

Figure 6-22

Constant ζ lines and
constant ω_n circles.

Figure 6–23

Constant ζ lines and
constant ω_n circles
superimposed on a
root-locus plot.

Figure 6–25
Root-locus plot of
conditionally stable
system.

less than $1/T_a$.

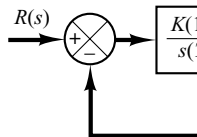


Figure 6–26

(a) Nonminimum-phase system;
(b) root-locus plot.

Orthogonality

feedback system where
the loci of $|G(s)H(s)|$
sponding to $\angle G(s)H(s)$

Section 6-3 / Plotted

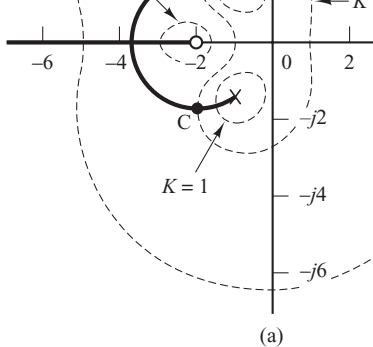


Figure 6–29

Plots of root loci and constant-gain $H(s) = 1$; (b) system with $G(s) =$

Root Loci for Positive-Feedback Systems

may be a positive-feedback system, the system may be stabilized by the outer feedback loop. The feedback inner loop is

The characteristic equation of the inner loop is

* Reference W-4

Section 6-4 / Root-Locus Plots of Positive-Feedback Systems

Rule 2 is Modified as follows:
of a test point on the

Rule 3 is Modified as follows:

An

where $n = \text{number of poles}$
 $m = \text{number of zeros}$

Rule 5 is Modified as follows:
rival) from a complex
of all angles of the vector
plex zero) in question

plex pole at $s = -\zeta\omega_n$

or

(The angle of

- 6.** Choose a test
apply the ang
angle conditio

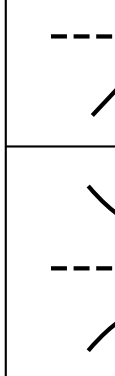
Figure 6–31 show
are shown with dash

Note that if

Figure 6–32

Root-locus plot for the
negative-feedback
system with

$$G(s) = K(s + 2)/[(s + 3)(s^2 + 2s + 2)],$$
$$H(s) = 1.$$



Heavy lines
curves corre

Section 6-4 / Root

series with the plan

An alternative t
ment(s) and place a
6–33(b). Such comp

In compensating
suitable design of a
sation and parallel
the power levels at v
nomic consideration

In general, series
series compensation
to provide isolation.
lowest energy point
ponents required in

(the output has a phase lag) and both phase lag and magnitude reduction occur in the low-frequency region. A compensator network is called a lag compensator.

Among the many types of lag compensators, lead compensators, lag compensators, and (tachometer) compensators are the most common types. Lead, lag, and tachometer compensators are used using operational amplifiers, relays, pneumatic, hydraulic, or combination of these.

Frequently used compensators are lead compensators, lag compensators, PID compensators. PID compensators are discussed in Chapter 7.

Section 6-5 / Root-Locus

Figure 6–35

(a) Root-locus plot of a three-pole system; (b), (c), and (d) root-locus plots showing effects of addition of a zero to the three-pole system.

Figure 6–36

Electronic circuit
that is a lead network
if $R_1C_1 > R_2C_2$ and a
lag network if
 $R_1C_1 < R_2C_2$.

○
 $E_i(s)$
○

in the complex plane
pensator in cascade

The procedures
6–38 by the root-loc

1. From the perf
inant closed-lo

Figure 6–38
Control system.

of these remaining
they are located near

EXAMPLE 6–6

Consider the position
function is

The root-locus plot for
for the system is

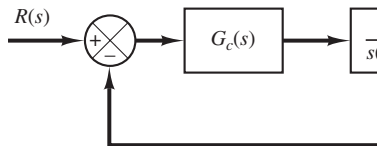


Figure 6–40

(a) Compensated system; (b) desired closed-loop pole location.

(a)

Figure 6-41

Determination of the
pole and zero of a
lead network.

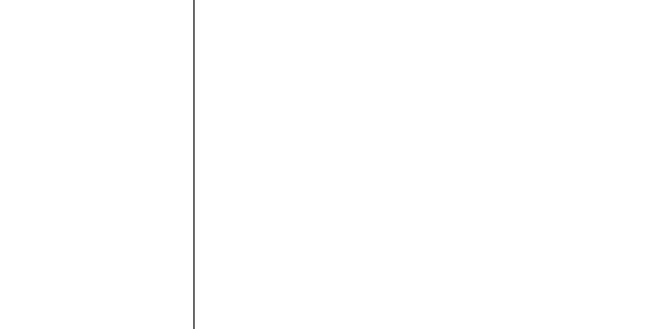


Figure 6–42
Determination of the
pole and zero of the
lead compensator.

the desired points in the s -plane. The zero at -1.9432 . Then
Since no restriction has been
given concerning the value of K_c ,
sign is satisfactory.

Method 2. If we place a
plant pole at $s = -1$,
Hence the lead compensator

The value of K_c can be

Section 6-6 / Lead

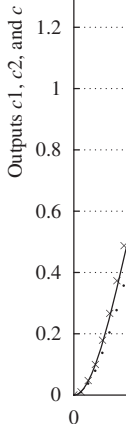
Notice that the system is type 1 and the steady-state error is constant. This means that for a constant reference input, the following ramp inputs the error will be zero.

For different combinations of K_p and K_v , the value of K_v will be different. This will affect the system's response to a ramp input. Changing the pole-zero location will affect the system's response to a step input. Then we must alter the system's response to a ramp input.

Comparison of steady-state error

In what follows we shall compare the steady-state error of the original uncompensated system with the steady-state error of the compensated system. Method 2. The MATLAB

Figure 6–45
Unit-step response
curves of designed
systems and original
uncompensated
system.



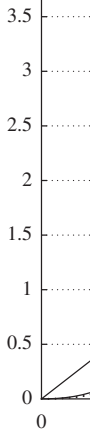


Figure 6–46
Unit-ramp response
curves of designed
systems.

the problem of finding a compensator that exhibits satisfactory characteristics. Compensating for the loss of loop gain without appreciably changing that the root locus is not a simple task. The root locus can be changed appreciably by adding a zero. This can be accomplished by adding a feedforward transfer function.

To avoid an appreciable change in the network should be made. The root locus is the pole and zero of the open-loop transfer function in the s plane. Then the closed-loop poles move slightly from their original positions. The root locus is changed only slightly.

Section 6-7 / Lag Compensation

loop transfer function
error constant K_v of

If the compensator
system with the open
 $\hat{K}_v$ becomes

where K_v is the static
Thus if the comp
constant is increase

6. Draw a new K so that the dominant closed-loop poles are in the desired location. The new work is very similar to the previous work, almost identical. You may want to locate, on the real axis, the transient-response time constant.
7. Adjust gain $\hat{K}$ so that the dominant closed-loop poles are in the desired location.

Figure 6–47
Control system.

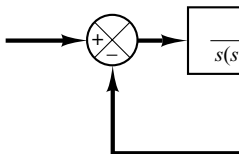


Figure 6–48
 (a) Control system;
 (b) root-locus plot.

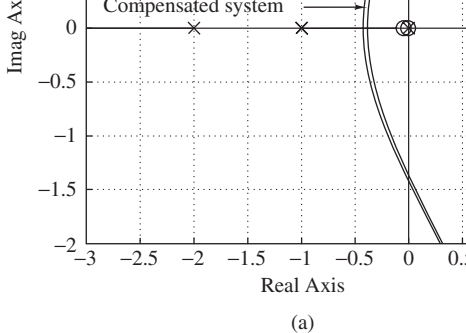


Figure 6–50

(a) Root-locus plots of the compensated system near the origin.

```
rlocus(num  
v = [-0.6 0  
grid  
title('Root-L
```

If the damping ratio
are obtained from the

The open-loop gain K

The undamped natural frequency of the system is 0.631 rad/sec. Therefore, we can see that the transient response of the compensated system will take less time to settle than the uncompensated system. The response will take less time to settle because the bandwidth will increase in the compensated system. The bandwidth will increase because the compensation as discussed in the next section.

Next, we shall compare the transient response of the uncompensated system with the transient response of the compensated system to a unit step input.

To obtain the unit step response of the compensated system, we need to find the transfer function $C(s)/[sR(s)]$. Since $C(s) = \frac{C(s)}{sR(s)}$, we can write the transfer function as

$$\frac{C(s)}{sR(s)}$$

Section 6-7 / Lag Compensation

```

den = [1 3 2
% ***** Specific
% step comman

t = 0:0.1:50;
c1 = step(num
c2 = step(num
plot(t,c1,'-',t,c
grid
text(2.2,27,'Co
text(26,21.3,'U
title('Unit-Ram
xlabel('t Sec');
ylabel('Output

```

```

den = [1 -5 2];
% ***** Spec
% step comm

t = 0:0.1:40;
c1 = step(num,den,t);
c2 = step(num,den,t);
plot(t,c1,'- ',t,c2,'- ');
grid
text(13,1.12,'Unit-Step')
text(13.6,0.88,'t = 0.1 sec')
title('Unit-Step Response')
xlabel('t Sec')
ylabel('Output')

```


Lead compensation system. Lag compensation reduces the speed of the response.

If improvement in the transient response is required, then both a lead compensator and a lag compensator are used. This is better than introducing both a lead and a lag compensator. However, it is economically not feasible.

Lag–lead compensation
Since the lag–lead compensator increases the order of the system, it increases the order of the system in the compensated system.

Electronic Lag–lead compensation
shows an electronic circuit for lag–lead compensation.

$$E_i(s)$$

The sign inverter ha

Thus the transfer fu

$$\frac{E_o(s)}{E_i(s)} = \frac{E_o(s)}{E(s)}$$

Let us define

$$T_1 = (R_1 + R$$

Section 6–8 / Lag-I

3. Assuming that
portion

Figure 6–54
Control system.

(The preceding

Case 2. $\gamma = \beta$.

procedure for the la

1. From the give
dominant clos

Section 6–8 / Lag-I

The value of βT_2 , the
too large to be phys
sator when $\gamma = \beta$ is

choose the zero at $s = -1$. When the zero is chosen, the calculation or graphical portion of the lag-lead

Figure 6–55
Control system.

Since $T_2 = 5$ satisfies

Now the transfer func

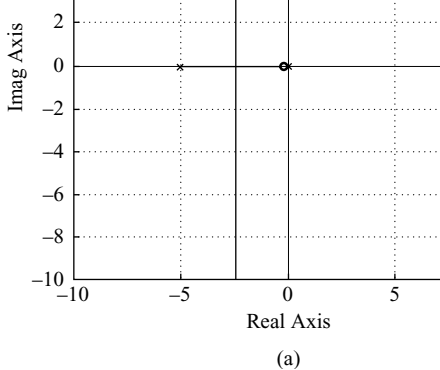


Figure 6–56

(a) Root-locus plot of the compensated system

system. (a) Unit-step
response curves;
(b) unit-ramp
response curves.

The unit-step responses
are shown in Figure 6-10
for the uncompensated and
compensated system.

EXAMPLE 6-9

Consider the control system
of the form given by Eqs. (6-1)

or

The phase-lead portion

For the phase-lag

$$\left| \frac{s + \frac{1}{T_2}}{s + \frac{1}{3.503}} \right|$$

Section 6–8 / Lag-I

The compensated system

No cancellation occurs. The angle contribution of the closed-loop poles are $\pm 90^\circ$. The closed-loop poles can be found from the characteristic equation of the compensated system

$$(s + 2)(s^2 + 8.8685s + 23.3)$$

which can be simplified to

$$\begin{aligned}s^4 + 8.8685s^3 + 23.3s^2 + 46.6s + 46.6 \\ = (s + 2.453)(s^3 + 6.4155s^2 + 18.8465s + 19.1465)\end{aligned}$$

Outputs
2.
1.
0.

Figure 6–59
(a) Unit-step
response curves for
the compensated and
uncompensated
systems;
(b) unit-ramp
response curves for
both systems.



Figure 6–60

(a) Series
compensation;
(b) parallel or
feedback
compensation.

EXAMPLE 6–10

Consider the system shown in Figure 6–61. Determine the range of k such that the system is stable. Here the system is

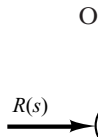


Figure 6–61
Control system.

Thus, the angle of departure for the system. Notice that the root terminates on the zero and terminates on the

Note that the closed-loop origin and making the intersections of the root locus. Thus, two values of K are the value of K is

K

Hence

Thus, we have two solutions
located at

$$s = -1.0$$

For $k = 1.4130$, the third

$$s = -2.5$$

It is important to note that
closed-loop zero. This means
have a closed-loop zero.


```
num1 = 1
den1 = 1
num2 = 2
den2 = 1
t = 0:0.01:10
c1 = step_response(num1,den1,t)
c2 = step_response(num2,den2,t)
plot(t,c1,'b',t,c2,'r')
text(2,0.5,'c1')
text(3,0.5,'c2')
grid
title('Unit Step Response')
xlabel('Time (s)')
ylabel('Amplitude')
```

ues of K it is underdetermined.

Solution. The procedure is:

1. Locate the open-circuit poles on the real axis between 0 and ∞ .
2. The number of open-circuit poles and zeros are no asymptotes.
3. Determine the breakaway point.

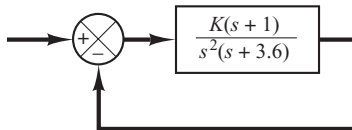
or

Example Problems and Solutions

Notice that both
points. At point s

Similarly, at $s = -$

(Because point s
 $s = -2.366$ lies b



(a)

Figure 6–65

(a) Control system; (b) root-locus

Example Problems and

The intersection of the

Next we shall find the

we have

or

Thus, the breakaway occurs at a double root. When a double root occurs, the value of gain K at point σ_{br} is

This means that with K at this value, the system can be easily verified

Example Problems and

Taking tangents of bo

we obtain

which can be simplifie

for small or large values of ζ , the system is underdamped or overdamped.

Solution. A root locus plot is shown in Figure 10.10. The poles of the root-locus equation are at $s = -1$ and $s = -2$.

Angle

The intersection of the root locus with the imaginary axis is at $s = -1$.

The breakaway and breakin points are at $s = -1.5$ and $s = -2.5$.

Example Problems and Solutions

the value of η is 2, and
The angle of departure

or

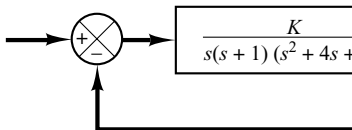
The root-locus branches
at $s = -1.6667$.

Next we determine
tuting $s = j\omega$ into the

or

from which we obtain

ω



(a)

Figure 6–68

(a) Control system; (b) root-locus

Example Problems and

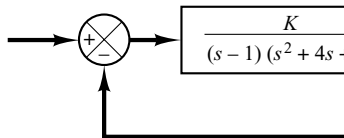
by substituting $s = j\omega$

or

from which we obtain

$$\omega =$$

The root-locus branches intersect at $\omega = \pm 1.6125$. Also, the root locus is on the imaginary axis for $\omega = \pm 1.6125$. Figure 6–68(b) shows the root locus for $\omega = \pm 1.6125$ that extends to the right.



(a)

Figure 6–69

(a) Control system; (b) root-locus

Example Problems and

we have

$$-\angle s -$$

By substituting $s = \sigma$

$$\angle \sigma - 1 + j\omega$$

or

$$\angle \sigma + 2 + j(\omega$$

which can be rewritten

$$\tan^{-1}\left(\frac{\omega +}{\sigma +}\right)$$

Solution. We shall plot the transfer function is given by

the equation for the a

$$\lim_{s \rightarrow \infty}$$

the equation for the a

Example Problems and

```

K1 = 0.0;
K2 = 0.3;
K3 = 0.5;
K4 = 10;
K = [K1 K2 K3 K4];
r = rlocus(N,K);
a = rlocus(N,K1);
y = [r a];
plot(y,'-')
v = [-4 4];
grid
title('Root Locus')
xlabel('Real Axis')
ylabel('Imaginary Axis')
% ***** M

```

```
K3 = 0.5  
K4 = 10  
K = [K1  
r = rlocu  
a = rloc  
plot(r,'o'  
hold  
Current  
plot(a,'-'  
v = [-4  
grid  
title('Ro  
xlabel('F  
ylabel('I
```

Example Problems and

where we used the fol

The expression

gives the equation for

Figure 6-72

Plot of root loci and asymptotes.

Example Problems and

```

num = [1 4 0]
den = [1 4 0]
K1 = 0:0.1:2
K2 = 2:0.0.2
K3 = 2.5:0.5
K4 = 10:1:5
K = [K1 K2 K3 K4]
r = rlocus(num,den,K)
plot(r, 'o')
v = [-8 2 -5]
grid
title('Root-Locus')
xlabel('Real')
ylabel('Imag')

```

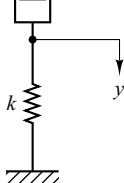


Figure 6–74
Mechanical system.

This is the transfer function

we obtain

This mechanical system

Example Problems and

[Note that depending on the choice of β , Equation (6–30) may or may not be satisfied.]

If such a β exists and $\beta > 0$, then the control system is asymptotically stable. If $\beta = 0$, the control system is marginally stable. If $\beta < 0$, the control system is unstable.

then the mechanical system is asymptotically stable.

Figure 6–77

Determination of the
pole of the lead
network.

Example Problems and

Figure 6–78
Root-locus plot of
the compensated
system.

Second Attempt: To contribute half the location of the zeros at $s = -4$ may be made.

Once we choose α as shown in Figure 6—

which yields

Example Problems and

A root-locus plot for the closed-loop zero near $s = -1$ in Figure 6–81(b).

The closed-loop transfer function is

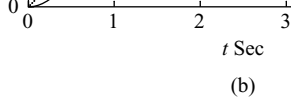
$$\frac{C(s)}{R(s)}$$

The closed-loop poles are at $s = -1$ and $s = -2$.

Figure 6–81
(a) Root-locus

Figures 6–82(a) and (b) show the root-locus for the system. The unit-step response is shown in Figure 6–82(c). Notice that in the unit-step response, the system has a feedback path of 1. The unit-step response is shown together with the unit-step response in Figure 6–82(c).

Example Problems and



- A-6-14.** Consider a system with the transfer function $G(s)$. Using the root-locus approach, design a proportional controller K_p (and T_d) such that the closed-loop system has a damping ratio ζ and a natural frequency ω_n is 0.5 rad/sec.

Solution. Note that the transfer function $G(s)$ has a pole at $s = -1$ and one zero at $s = -2$.

Since the desired closed-loop system has a damping ratio ζ and a natural frequency ω_n is 0.5 rad/sec, the desired closed-loop poles are located at

desired closed-loop p

The open-loop poles a
diagram shown in Fig

This means that the z
location of the zero as

Example Problems and

Solution. Assume that

R
—

Figure 6–84
Control system.

The closed-loop trans

To compare the transi
and unit-ramp respon
6–85(a) and (b), respe
6–85(c). The designed

Example Problems and

Figure 6–85

(a) Unit-step responses of the compensated and uncompensated systems; (b) unit-ramp responses of both systems; (c) unit-ramp responses showing steady-state errors.

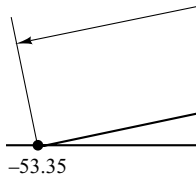
Figure 6–86

Root-locus plot of

$$G(s) = 10/[s(s + 2)(s + 8)].$$

Example Problems and

Figure 6–87
Graphical
determination of the
zero and pole of
the lead portion
of the compensator.



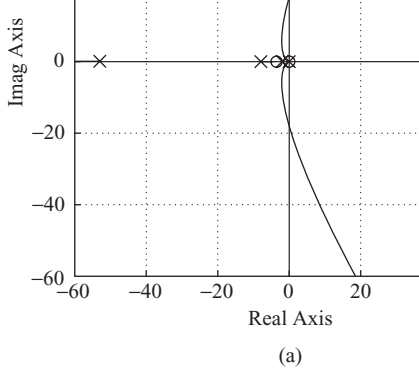


Figure 6–88

(a) Root-locus plot of compensated system

Example Problems and

compensated and uncompensated systems; (b) unit-ramp responses of both systems.

To verify the impulse responses and unit-ramp responses, compare Figures 6–89 (a) and (b).

- A–6–17.** Consider the system whose static velocity error constant K_v is 0.5. (Check that the pole at $s = -1$ is the only pole of the system.)

For the lag portion of

where $s = s_1$ is one of
tion of the compensat

$$G_c(s_1)G($$

Example Problems and

it can be found that

Thus

The lag portion of
and zero of the lag po

That is,

$$\frac{C(s)}{R(s)}$$

The closed-loop poles

Notice that the dominant closed-loop poles $s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$ are the dominant closed-loop poles. [The dominant closed-loop pole approximations involve (6-36) and (6-37).]

Example Problems and

system; (b) root-locus plot near the origin.

Figures 6–92(a) and (b) are the root-locus plots for the system, respectively. The root-locus plot in Figure 6–92(a) shows a real pole at $s = -0.16025$. However, this pole produces a long tail of small oscillations that extend much farther to the left than the real part of this real pole on the real axis. The root-locus plot in Figure 6–92(b) shows a complex pole at $s = -1.8308 \pm j3.235$. The characteristics of the system response following the unit-ramp

(b) unit-ramp
response of the
compensated system.

A-6-18. Figure 6-93(a) is a block diagram of a control system with the transfer function for the closed-loop system

$$\frac{C(s)}{R(s)} = \frac{K(s+1)}{s^2 + 2s + K}$$

The unit-step response of the system exhibits frequency oscillations. The response is dominated by the complex poles of the system.

Example Problems and Solutions

Solution. Figure 6–94 shows the root locus for the system with a zero at $s = -0.05$ and a pole at $s = -0.05$. Such a system has a response quite slow. Hence, the system is not from the $j\omega$ axis—for



Figure 6–94
Compensated
attitude-rate control
system.

—|—
-12

Figure 6–95
Pole and zero of
 $\hat{G}_c(s)$.

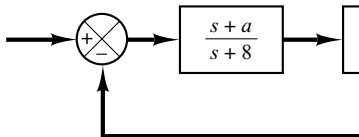
Example Problems and

Figure 6–96
Root-locus plot of
the compensated
system.

Figure 6–97

(a) Unit-step
response of the
compensated system;
(b) unit-ramp
response of the
compensated system.

Example Problems and



(a)

Figure 6–98

(a) Control system; (b) root-locus

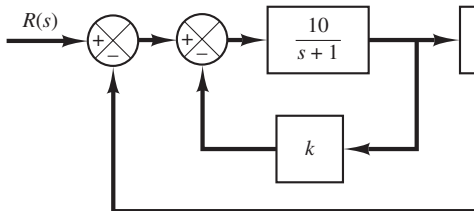
$$\omega =$$

The crossing points are the points where the root locus branch touches the imaginary axis for the system.

Since the damping ratio ζ is a measure of the closed-loop pole in the s plane, the root locus in the upper-half s plane is the desired dominant closed-loop pole location.

At these points, the value of ζ is zero.

Example Problems and Solutions



(a)

Figure 6–99

(a) Control system; (b) root-locus plot, where

Notice that $\omega = 0$ corresponds to $K \geq 0$

is an equation of a circle. This circle that lies to the left of the imaginary axis is the portion of the circle with $K < 0$. Figure 6–99(

Since we require the root locus and a line to intersect on the real axis. The intersection Hence, the desired va

Example Problems and

Locate the closed-loop poles on the root loci s. The dominant closed-loop poles have a damping ratio of 0.5. Determine the corresponding value of gain K .

B-6-7. Plot the root loci for the system shown in Figure 6-100. Determine the range of gain K for stability.

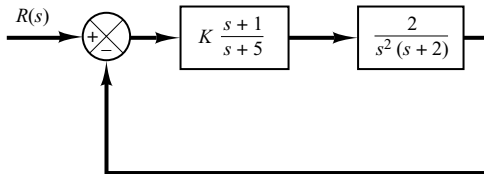


Figure 6-100
Control system.

Figure 6–103

Mechanical system.



B–6–14. Consider the system shown in Figure the root loci for the system. Determine the value that the damping ratio ζ of the dominant closed is 0.5. Then determine all closed-loop poles. Plot the step response curve with MATLAB.

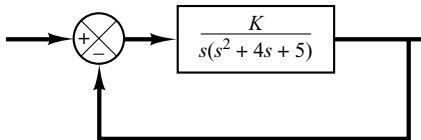


Figure 6–104 Control system.

Problems

Figure 6-110

Angular-positional system.

B-6-21. Consider the control system shown in Figure 6-111. Design a compensator such that the dominant poles are located at $s = -2 \pm j2\sqrt{3}$ and the steady-state error constant K_v is 50 sec^{-1} .

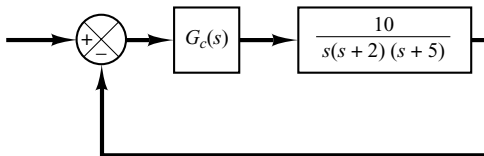


Figure 6-111

Control system.

Figure
Contr

Problems

its mathematical models and the methods that are employed. Control systems are designed by

Frequency-response methods were developed by Nichols, and many of the advantages of frequency-response control over conventional control theory are apparent.

The Nyquist stability criterion is a method for determining the stabilities of linear control systems from their frequency-response characteristics. It is a powerful tool that is readily available since the frequency-response characteristics of the transfer function of a system can be determined from the frequency-response characteristics of the system. It is a powerful tool that is readily available since the frequency-response characteristics of the transfer function of a system can be determined from the frequency-response characteristics of the system. It is a powerful tool that is readily available since the frequency-response characteristics of the transfer function of a system can be determined from the frequency-response characteristics of the system.

shift between the input and the output, and the input frequency.

The steady-state input does not depend on the initial condition.) If $Y(s)$ is the Laplace transform (7-1) when $x(t)$ is

$$Y(s) =$$

=

Figure 7-1

Stable, linear, time-invariant system.

expression for $G(-j\omega)$

Then, noting that

Equation (7-4) can

a lead network, while

EXAMPLE 7–1

Consider the system s

For the sinusoidal inp

Substituting $j\omega$ for s i

Figure 7–3

First-order system.

Section 7–1 / Intro

and

Thus the steady-state

y_{ss}

From this expression,
then the network is a

Presenting Fre
sinusoidal transfer f
its magnitude and
commonly used rep

Figure 7-4
Number-decibel
conversion line.

equal to 0 dB at $\omega = 1$.
If the transfer function has n poles, the magnitude becomes, respectively

$$20n \text{ dB}$$

or

The slopes of the magnitude asymptotes are $-20n$ dB/decade and the phase asymptotes approach $-90^\circ \times n$ over the entire frequency range ($0 \text{ dB}, \omega = 1$).

Section 7-2 / Bode

This is an approximation whose magnitude equals 0 dB. The asymptotic approximation is $-20 \log \omega T$ dB decreasing at -20 dB/decade. The curve is thus a straight line with slope -20 dB/decade.

Our analysis shows that the magnitude curve of the factor $1/(1 + j\omega T)$ is one a straight line at low frequencies and one a straight line at high frequencies. The line with slope -20 dB/decade. The exact log-magnitude curve is shown in Figure 7–6.

The frequency $\omega = 1/T$ is the *break* frequency. For frequencies $\omega < 1/T$, the asymptotic expression is 0 dB. For frequencies $\omega > 1/T$, the asymptotic expression is $-20 \log \omega T$ dB.

The error in the
 The maximum error
 since

$$-2$$

The error at the f
 $\omega = 1/(2T)$ —is

$$-20$$

The error at the freq
 is

$$-20$$

Section 7-2 / Bode

Figure 7-7

Log-magnitude error
in the asymptotic
expression of the
frequency-response
curve of
 $1/(1 + j\omega T)$.

dB

Figure 7-8

Log-magnitude
curve, together with
the asymptotes, and
phase-angle curve
for $1 + j\omega T$.

The high-frequency
frequency

This frequency, ω_n ,

The two asymptotic
frequency $\omega = \omega_n$,
The damping ratio
ously exist in the a
error depends on t
exact log-magnitud

$$\phi =$$

[

The phase angle is at
the corner frequency

At $\omega = \infty$, the phase
about the inflection
such phase curves. We

Section 7-2 / Bode

the minimum value
 ω_r is

As the damping
 $0 < \zeta \leq 0.707$, the
 $\omega_d = \omega_n \sqrt{1 - \zeta^2}$, v
it can be seen that f
creases monotonica
for all values of $\omega >$
the oscillations are

to plot $20 \log |G|$ in magnitude.
Here, however, we can do this
without using MATLAB.

First rewrite the transfer function as discussed above. Then plot the magnitude. Finally, draw the asymptotic approximation of the frequencies. The exact magnitude can be obtained by adding proper corrections.

The phase-angle plot can be obtained by summing the curves of individual components.

The use of Bode plots is much faster than other methods for determining the transfer function. The magnitude plot and the phase plot are added are the magnitude and phase plots.

Section 7-2 / Bode Plots

at the corner frequency. Correction
frequency. Correction
log-magnitude curve

Note that any ch
frequencies of the tra
curves and adding th
individual curves. We
is, the straight line wi
we get the effect of
 $\omega = \sqrt{2}$. The comple
-20 to -60 dB/decad
the slope to -80 dB/
change the slope from

For plotting the co
sketched. The algebra
as shown in Figure 7-

nonminimum-phase
are called *minimum*
functions are called

For systems with
minimum-phase tra
phase angle of any r

It is noted that f
determined from th
not the case. Multi
magnitude curve, b

Consider as an
respectively,

$$G_1(j\omega)$$

the system is minimum phase, the asymptote of the log-magnitude curve at $\omega = \infty$ is 0 dB, and the phase-angle at $\omega = \infty$ is 0°.

Figure 7-13

Phase-angle characteristics of the systems $G_1(s)$ and $G_2(s)$ shown in Figure 7-12.

Figure 7-14
Phase-angle
characteristic of
transport lag.

0

-100

-200

-300

Figure 7-15

Bode diagram for the
system $e^{-j\omega L}/(1 + j\omega T)$
with $L = 0.5$ and $T = 1$.

Figure 7-17
Log-magnitude curve
of a type 0 system.

Figure 7–18
Log-magnitude curve
of a type 1 system.

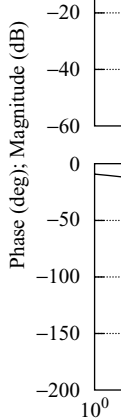
Figure 7-19
Log-magnitude curve
of a type 2 system.

To specify the 1
 (d1,d2,n).logspace(
 between decades 10
 between the endpo
 enter the command

Figure 7–20

Bode diagram of

$$G(s) = \frac{25}{s^2 + 4s + 25}.$$



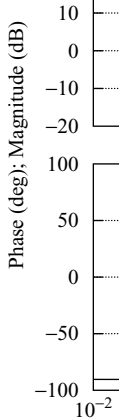


Figure 7-22

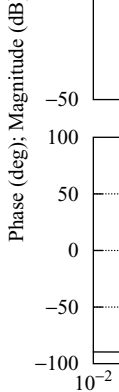
Bode diagram of

$$G(s) = \frac{9(s^2 + 0.2s + 1)}{s(s^2 + 1.2s + 9)}.$$

Figure 7-23

Bode diagram of

$$G(s) = \frac{9(s^2 + 0.2s + 1)}{s(s^2 + 1.2s + 9)}.$$



or

EXAMPLE 7-7 Consider the following

This system has one in

7-3 POLAR PLOTS

The polar plot of a system is a plot of the magnitude of the frequency response versus the phase angle. Thus, the polar plot is a plot of the magnitude of the frequency response versus the phase angle. Note that in the polar plot, the magnitude is plotted on the radial axis and the phase angle is plotted on the angular axis. The polar plot is a clockwise (clockwise) plot. An example of a polar plot is shown in Figure 7-3. The polar plot of $G(j\omega)$ represents the frequency response of the system. In the polar plot, it is important to note that the magnitude of $G(j\omega)$ on the real axis is the same as the magnitude of $G(j\omega)$ on the imaginary axis.

Section 7-3 / Polar

the values of $G(j\omega)$

If ω approaches infinity, $G(j\omega)$ approaches -90° . The magnitude of $G(j\omega)$ as the frequency ω is varied from 0 to infinity is shown in Figure 7.10. The point 0.5 on the real axis, $\angle = 0^\circ$, is the point where the locus crosses the real axis.

To prove that the locus is a circle, define

Quadratic Factor
frequency portions of

G

are given, respectively

lim
 ω

The polar plot of the
 ω increases from zero
to the negative real

Section 7-3 / Polar

system, the character
for sufficiently large
response becomes v

Figure 7–29
Polar plot showing
the resonant peak
and resonant
frequency ω_r .

Sketch a polar plot of

Since the sinusoid

G

the low-frequency po

and the high-frequen

Section 7-3 / Polar

Since the magnitude
monotonically and in
in Figure 7–32.

Figure 7–32
Polar plot of
 $e^{-j\omega L}/(1 + j\omega T)$.

Figure 7–33

Polar plots of type 0,
type 1, and type 2
systems.



Figure 7
Polar plot

Drawing Nyquist
are commonly used
feedback control s
rectangular plots. O
tion, but a given op

The MATLAB c
time, linear, time-in
produces a Nyquist

Section 7-3 / Polar

fied in the vector w
for each element in

EXAMPLE 7–10 Consider the following

Draw a Nyquist plot v
Since the system i

may be used to draw
in Figure 7–36. In thi
determined.

Or, combining these two

See MATLAB Program

Section 7-3 / Polar

Figure 7–38
Undesirable Nyquist
plot.

Figure 7-39

Nyquist plot of

$$G(s) = \frac{1}{s(s + 1)}.$$

Drawing Nyquist

system defined by

where $\mathbf{x}$ = state vector

$\mathbf{y}$ = output

$\mathbf{u}$ = control

$\mathbf{A}$ = state matrix

$\mathbf{B}$ = control matrix

$\mathbf{C}$ = output matrix

$\mathbf{D}$ = direct transmission matrix

MATLAB Program 7
using either of these two
Program 7–9.

Section 7–3 / Polar

MATLAB Program 7

curve, while in the
gram are combined
plot can easily be co
from the Bode diag
the gain constant of
decreasing gain), bu

Advantages of t
the closed-loop syst
out easily.

The log-magnitu
that for $1/G(j\omega)$ are

Section 7-4 / Log-M

in three different regions. The distance between the peak value of $G(j\omega)$

Since log-magnitude has been discussed in detail, examples of some log-magnitude plots, ever, more on Nichols

-20
-180°

7-5 NYQUIST STABILITY CRITERION

The Nyquist stability criterion is a method for determining the stability of a closed-loop system from the open-loop frequency response.

This section presents the Nyquist stability criterion. Consider a system with open-loop transfer function $G(s)$.

Section 7-5 / Nyquist

We shall show that, through any singular number and direction curve play a particular number and direction

Consider, for ex

The characteristic e

tour in the s plane encloses
 the $F(s)$ plane by the
 the contour in the s plane
 es neither the zero nor the
 plane by the locus of the
 poles.

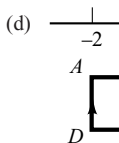
From the foregoing, we can
 gin of the $F(s)$ plane. If the
 encloses a pole or a zero, the
 in the right-half or left-half
 a pole or zero does not
 zeros, then the corresponding
 gin of the $F(s)$ plane. This is
 theorem, which is the basis of

[#]A complex function $F(s)$ is analytic in the s plane if it is differentiable at every point in the plane.

Section 7-5 / Nyquist Criterion

Figure 7-46

Closed contours in the s plane and their corresponding closed curves in the $F(s)$ plane, where $F(s) = (s + 1)/(s - 1)$.



Mapping Theorem

Let N be the number of poles and Z be the number of zeros of $F(s)$ in the s plane, with $N \neq Z$. Let C be a closed contour in the s plane, with N encirclements of the poles and Z encirclements of the zeros, that it does not pass through any poles or zeros. The contour C is then mapped into the $F(s)$ plane, where it becomes a closed curve with N encirclements of the origin. The mapping theorem states that the number of encirclements of the origin in the $F(s)$ plane is then $N - Z$. In other words, the number of encirclements of the origin in the $F(s)$ plane is the difference between the number of poles and the number of zeros of $F(s)$ in the s plane.

the function of $1 +$
radius. Because of t
 $1 + G(s)H(s)$ plan
in the s plane—that

Figure 7-47
Closed contour in
the s plane.

however, that a similar
of this is not given here
from the open-loop transfer
elements of the -1
function is a ratio of

Nyquist Stability
the $-1 + j0$ point by
stability criterion:

Nyquist stability
zeros on the $j\omega$
tion $G(s)H(s)$
then for stability
 $-1 + j0$ point

Figure 7-49
Region enclosed by a
Nyquist plot.

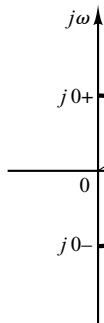


Figure 7–50

Contour near the origin of the s plane and closed contour in the s plane avoiding poles and zeros at the origin.

Then

As θ varies from -90° to -180° , as shown in Fig. 7-10, the locus encircles the -1 point twice, so that there are two zeros of $1 + C$ in the right half-plane, and the system is unstable.

Section 7-5 / Nyquist

wise direction. (Neg

In examining the
 rion, we see that the

1. There is no en
 ble if there are
 tem is untabl
2. There are one
 case the syste
 same as the nu
 system is unst
3. There are one
 the system is u

In the following
 stants (such as T , T_1

Figure 7-54
Polar plots of the
system considered in
Example 7-15.

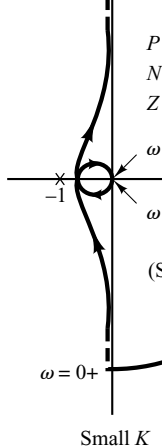


Figure 7–55
Polar plots of the
system considered in
Example 7–16.

$T_1 < T_2$
(Stable)

EXAMPLE 7–17 Consider the closed-loop

Determine the stability

Figure 7-57
Polar plot of the
system considered in
Example 7-18.



Figure 7–59
Multiple-loop
system.

the right-half s plane.

Once we find the root locus of $G(s)$, where

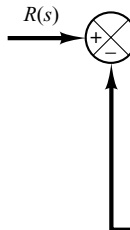


Figure 7-60
Control system.

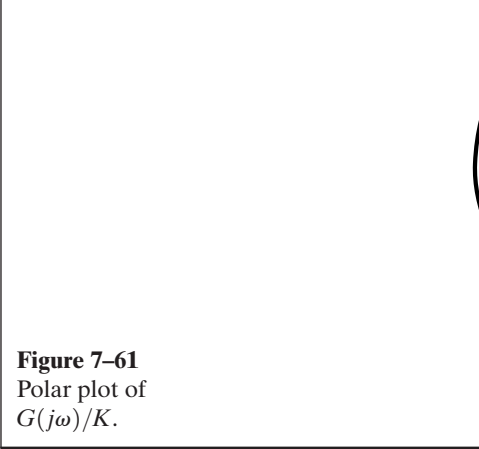


Figure 7-61
Polar plot of
 $G(j\omega)/K$.

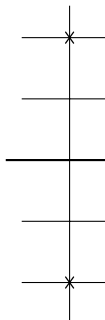
then the number of
comes infinite, and the
plot of such an oper

In general, if ex
form, both the $G(j\omega)$
the number of right
difficult to determine
mine whether a gi
right-half plane po
ponent is stable).

Figure 7–62

Modification of a
system with feedback
elements to a unity-
feedback system.

Figure 7-64
Two systems with
two closed-loop
poles each.



(a)

Figure 7-66

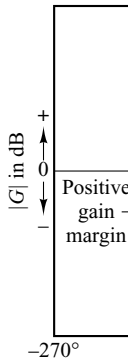
Polar plots of

$$\frac{K(1 + j\omega T_a)(1 + j\omega T_b) \cdots}{(j\omega)(1 + j\omega T_1)(1 + j\omega T_2) \cdots}.$$

Figure 7-67

Phase and gain margins of stable and unstable systems.

- (a) Bode diagrams;
- (b) polar plots;
- (c) log-magnitude-versus-phase plots.



more phase crossover
numerator dynamic
in Figure 7–68. For
phase margin is me

A Few Comme

a control system are
Therefore, these ma

It should be note
not give a sufficien
determination of re

For a minimum-
the system to be sta

Proper phase an
and are specified fo

the system is stable,
frequency is -60 dB

For nonminimum
quires careful study
tems is to use the N

EXAMPLE 7-20

Obtain the phase and
 $K = 10$ and $K = 100$

Figure 7-69
Control system.

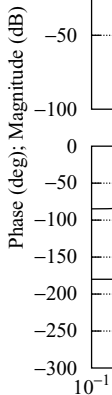
Section 7-7 / Relat

Thus, the system is sta

Notice that one of which the effects of gas must increase the phase in K is not desirable. This suggests that reshaping be necessary. Components

Obtaining Gain Crossover Frequency, and Gain-Mand to be used is

Figure 7-72
Bode diagram of
 $G(s)$ shown in
Figure 7-71.



closed-loop poles with a smaller response. A smaller of a pair of dominant system is well damped.

Remember that if $\zeta > 0.707$. [The values of M_r and for checking agree

Figure 7–73
Standard second-order system.

the phase angle of C

$$\angle G(j\omega) =$$

Thus, the phase margin

Equation (7-21) gives
 γ . (Notice that the p

Section 7-7 / Relat

indicative of the

3. From Equation (7.10), it is seen that for a larger the value of ζ is shown in Figure 7.10. It is seen for $\zeta > 0$, the value of M is

Correlation between

General Systems.

of the frequency response of the system. The approach compared with the system to aperiodic inputs that is of primary interest and frequency response

For linear, time-invariant, second-order, conjugate closed-loop systems, the transient response can be characterized by the following two properties:

1. The value of the steady-state response (performance) is unity (0 dB) for $M_r = 1$ ($0 \text{ dB} < M_r < 1$), and the system exhibits severe undershoot for $M_r > 1$ ($0.4 < \zeta < 0.7$). The system exhibits severe overshoot for $M_r < 1$ ($0.4 < \zeta < 0.7$), and the response to a large overshoot is due to the noise signals which are amplified in the system.
2. The magnitude of the transient response (rise time) is inversely proportional to the rise time. In other words, the rise time is inversely proportional to the magnitude of the transient response.

Section 7-7 / Relative Stability

creases with increase
the increase in ζ . The
each other.

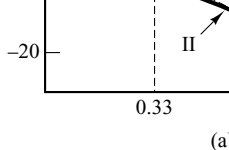
Figure 7-76

Plot of a closed-loop
frequency response
curve showing cutoff
frequency ω_b and
bandwidth.

Figure 7-77

Comparison of dynamic characteristics of the two systems considered in Example 7-22.

(a) Closed-loop frequency-response curves; (b) unit-step response curves; (c) unit-ramp response curves.



tem, the closed-loop
loop frequency resp
The closed-loop tra

In the Nyquist or po
where ω_1 is the freq
angle of the vector
to the Nyquist locu
sents the closed-loo

Section 7-8 / Close

where X and Y are

and M^2 is

Hence

λ

If $M = 1$, then f
straight line parallel



Figure 7-81

A family of constant
 M circles.

This is an equation
 $\sqrt{\frac{1}{4} + 1/(2N)^2}$. For
the radius of the circle
respectively. Since M
regardless of the value
The constant α loci of
 N circles is shown in

It should be noted
entire circle, but only
of the same circle. The
(or multiples thereof)

The use of the M
response from the circle
tude and phase of the

magnitude M_r . If it
then the system sho
time, there should b

Figure 7–83(a) s
7–83(b) shows the C
it is possible to obta
the $M = 1.1$ circle
that at this frequen
ure 7–83(a), the M
point on the $G(j\omega)$ l
closed-loop frequen
frequency ω curve, a

The resonant pe
est radius that is tar

Section 7–8 / Close

(b) $G(j\omega)$ locus
superimposed on a
family of N circles;
(c) closed-loop
frequency-response
curves.

peak value M_r and the
to the $G(j\omega)$ locus.

Nichols Chart.

the M and N loci in
 M and N loci in the
The $G(j\omega)$ locus dr

Figure 7–84
Nichols chart.

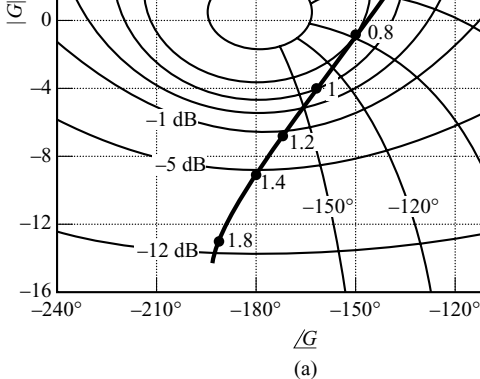


Figure 7–85

(a) Plot of $G(j\omega)$ superimposed on Nichols chart

Figure 7-86
Determination of the
gain K using the
Nichols chart.

changes from -20 to 0 dB/decade. If the corner frequency exists in the transfer function, the magnitude plot must be a quadratic curve.

in the transfer function. The magnitude plot is equal to the corner frequency. The magnitude plot is experimentally obtained. The magnitude plot is a peak near the corner frequency. Figure 7–9.

Once the factors are determined, the magnitude plot can be determined from the transfer function.

which this asymptote approaches
to $\sqrt{K}$.

Examples of log magnitude and phase plots are shown in Figure 7-87, together with the asymptotic approximations.

The experimental transfer function obtained from the experimental plot of the magnitude and phase-angle curve or from the experimental corner frequencies) should be compared with the theoretical transfer function obtained from the numerator and denominator of the transfer function model.

Section 7-9 / Experimental Determination of Transfer Function

constant rate of change of the
the transfer function

where $G(s)$ is a rational function

$$\lim_{\omega \rightarrow \infty} \frac{1}{\omega} \frac{dG(j\omega)}{d\omega}$$

where we used the fact that
can evaluate the magnitude

The value of the damp
Referring to Figure 7–
at the intersection of th
the 0-dB line. The valu

or

Section 7–9 / Exper

transport lag.

Hence, we assume
between the computed a
determine the value c

or

The presence of trans
mined from the exper

nents are usually determined experimentally obtaining the magnitude plots when the Bode frequency noises were used in other approaches.

There are basically two plot approach and added, the polar plot. A new polar plot, which is a diagram of the complex number, thus plotting the complex number. As the frequency is varied, the magnitude curve, and the phase curve, and the phase curve to work with the Bode

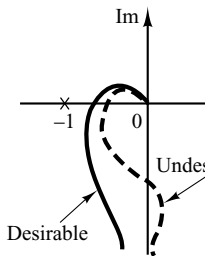


Figure 7–89

(a) Examples of desirable root locus
 (b) examples of undesirable root locus

We shall first examine
present a design technique

Characteristics
following transfer function

where α is the attenuation
and a pole at $s = -\alpha$
the right of the pole
cated far to the left

Section 7-11 / Lead

dB

-1

-2

90

Figure 7-92

Bode diagram of a
lead compensator

$$\alpha(j\omega T + 1)/(j\omega\alpha T + 1),$$

where $\alpha = 0.1$.

Determine gain

2. Using the gain adjusted but u
3. Determine the additional 5° t

Figure 7-93
Control system.

The compensated system
Define

where $K = K_c \alpha$.

Figure 7–94
Control system.

Figure 7-95

Bode diagram for

$$G_1(j\omega) = 10G(j\omega) \\ = 40/[j\omega(j\omega + 2)]$$

Thus, the transfer function

Note that

The magnitude curve of the compensated system has

crease in this frequency response. The phase a The compensated sys relative-stability requ

Note that for type ity error constant K_v the extension of the in Note also that we have from -40 dB/decade



Figure 7-97
Compensated
system.

Figure 7–98

Polar plots of the gain-adjusted but uncompensated open-loop transfer function G_1 and compensated open-loop transfer function $G_c G$.

```
grid
title('Unit-Ram
xlabel('t Sec')
ylabel('Output
text(0.89,3.7,'C
text(2.25,1.1,'U
```

It is noted that the

Because the dominant
quickly.

Section 7-11 / Lea

7-12 LAG COMPENSATION

In this section we first
Then we present lag

Characteristics
following transfer f

quencies. Thus, the

Lag Compensa

The primary function of the frequency range to is of no consequence

The procedure for
by the frequency-re

1. Assume the fo

C

below the new
Notice that
the phase lag of
margin.

4. Determine the
at the new gain
termine the v
pole of the lag
5. Using the valu
culate constan

With $K = 5$, the com
We shall next plot

Figure 7–103
Control system.

compensator. The required transfer function is $-12/s$ to be 0.5 rad/sec . To bring the lag compensator must

or

The other corner frequency of the compensator, is then determined

Figure 7–105

Log-magnitude-versus-phase plots of G_1 (gain-adjusted but uncompensated open-loop transfer function) and $G_c G$ (compensated open-loop transfer function).

```

num1 = [1],
den1 = [0.5 1]
num1c = [50]
den1c = [50 1]
t = 0:0.1:20;
[y1,z1,t] = step
[y2,z2,t] = step
plot(t,y1,'.',t,y2)
grid
title('Unit-Ram
xlabel('t Sec')
ylabel('Output
text(8.3,3,'Con
text(8.3,5,'Unc

```

compensated and
uncompensated
systems (Example
7-27).

Note that the zero and

Zero at

Poles at

The dominant closed-loop
is slow. Also, a pair of
slowly decreasing tail

Section 7-12 / Lag

4. Since the lag compensator is designed as a proportionally-integrating system, the system tends to be more stable. The time constant T should be chosen such that the system is stable.
5. Conditional stability: A system is conditionally stable if it is stable for some values of the gain K and unstable for other values. A system is unconditionally stable if it is stable for all values of the gain K . To avoid this, the system should be made less stable and the gain K should be chosen such that the system becomes stable. This is done by adding a lag compensator to the system.

where $\gamma > 1$ and β

produces the effect

produces the effect

Section 7-13 / Lag

dB -10

-20

-30

90

0

-90

0

Figure 7-110

Bode diagram of a
lag-lead
compensator given
by Equation (7-27)
with $K_c = 1$,
 $\gamma = \beta = 10$, and
 $T_2 = 10T_1$.

Hence,

We shall next draw the phasor diagram for Figure 7-111. The phase angle is -32° , which indicates

The next step in the design is to determine the required frequency. From the phase margin, the required frequency can be determined. It is convenient to choose the frequency such that the required lead angle required at the frequency is equal to the phase margin of the lag-lead network.

Once we choose the frequency of the phase frequency $\omega = 1/T_2$ (which be 1 decade below the

$\beta = 10$. (Note that we

Then the corner frequency of the compensator) becomes the lag-lead compensator.

The phase-lead portion of the compensator is $\omega = 1.5$ rad/sec, and the lag portion of the compensator contributes the required phase lead. From this requirement, the point on the -20 -dB line determines

Figure 7-112

Polar plots of G (gain adjusted) and $G_c G$.

Figure 7-114

Unit-ramp response
of the compensated
system (Example
7-28).

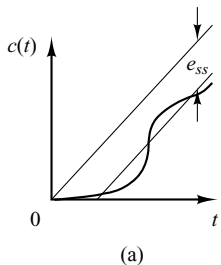
2. In some designs, the lead compensator satisfies the specification with less gain than is possible with a lag compensator. This means a larger bandwidth. The bandwidth is increased with lag compensation, but the lead compensator provides a larger bandwidth. The lead compensator is more vulnerable to noise since the lead compensator provides less compensation at high frequencies.
3. Lead compensation is often used in conjunction with lag compensation. The lag compensation provides a larger gain than the lead compensator, but the lead compensator implies larger bandwidth.



Figure 7–115

Unit-step response curves and unit-ramp response curves.

(a) Uncompensated system; (b) lead compensated system; (c) lag compensated system; (d) lag–lead compensated system.

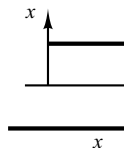


It is noted that canceled, as is almost the response to have a dominant. If the cancellation be small.

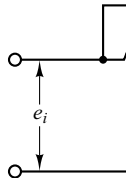
It should be noted that the function of unity. It

Figure 7-116

Step-response curves showing the effect of canceling a large time constant.



Section 7-13 / Lag



$$\frac{E_o(s)}{E_i(s)} = \frac{1}{RC_1 s + 1}$$

Figure 7-117
Bridged- T networks.

Figure 7-118

Bode diagram for

$$10(1 + j\omega)/[(2 + j\omega)(5 + j\omega)].$$

Example Problems and

Assuming that $U_2(j\omega)$

Similarly, assuming th

Notice that $Y_2(j\omega)/U_2$

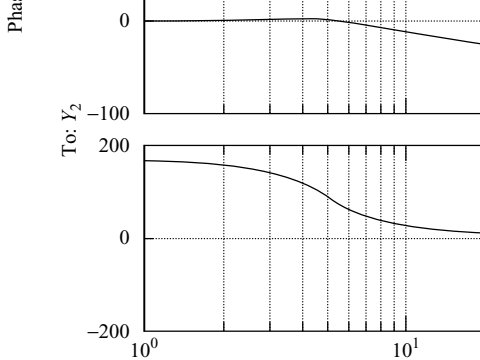


Figure 7-119
Bode diagrams.

Example Problems and

```

grid
title('Bode Diagram')
xlabel('Frequency')
ylabel('Gain (dB)')
text(1.2,-31,'K = 1')
text(1.1,-8,'K = 10')
text(11,-31,'K = 20')
semilogx(w,phase)
grid
xlabel('Frequency')
ylabel('Phase (deg)')
text(0.2,-90,'K = 1')
text(0.2,-20,'K = 10')
text(1.6,-20,'K = 20')

```

Figure 7-121

Bode diagrams:
(a) Magnitude-
versus-frequency
curves; (b) phase-
angle-versus-
frequency curves.

-250
-300
 10^{-1}

A-7-5. Prove that the polar plot

is a semicircle. Find the

Example Problems and

This may be seen from

then $\hat{F}(s)$ has a zero at

Hence,

We see that by taking
 $\hat{F}'(s)/\hat{F}(s)$.

[#]For the definition of an

$$\oint \frac{F'}{F}$$

The integral $\oint_F d \ln |F|$
 point of the contour L

The angular difference
 the phase angle of F'
 contour. Noting that L
 and $\theta_2 - \theta_1$ is zero or

Example Problems and

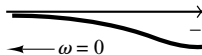


Figure 7–123

(a) Nyquist plot;
 (b) complete Nyquist
 plot in the G plane.

Substituting $\omega = 1/\sqrt{V}$

The critical value of the

Hence,

The system is stable if

Example Problems and

Figure 7-125

(a) Polar plot of $K/(j\omega - 1)$;
(b) polar plots of $K/(j\omega - 1)$ for stable and unstable cases.

$\omega = 0$



$$P = 1$$

$$N = -1$$

$$Z = 0$$

Figure 7-126
Polar plots of
 $2.65e^{-0.8j\omega}/(1 + j\omega)$
and $2.65/(1 + j\omega)$.

Example Problems and

$$G(s) = \frac{1}{s(s+1)(s+10)}.$$

A-7-12. Consider the same system as in Problem A-7-11, but with $\omega = 1$ rad/sec. Plot the magnitude and phase of the positive-frequency response.

Solution. Drawing a Bode plot of the following command:

The frequency response of the system is shown in Figure 7-10. For example, the frequency response at $\omega = 1$ rad/sec is

Nyquist plot for the positive-frequency region.

A-7-13. Referring to Problem

Locate on the polar plot
Also, find the magnitude

Solution. In MATLAB
three frequency sub

Example Problems and

```
% ----- The following table shows the magnitude and phase
% the corresponding to the frequency response of the system.
% [w mag phase]
ans =
    0.2000    4.9000
    0.3000    3.2000
    0.5000    1.9000
    1.0000    1.5000
    2.0000    1.7000
    6.0000    1.6000
   10.0000    1.4000
   20.0000    0.8000
```

feedback system:

MATLAB

```
num =  
den =  
nyquist  
grid  
title('N
```

Example Problems and

Notice that $1/G(s)$ has
Therefore, the Nyquist

reduces to $Z = N$ since
 $1 + [1/G(s)]$ in the r
the $-1 + j0$ point. For
ure 7-132 shows the N
Notice that since

Example Problems and

A-7-16. Figure 7-133 shows a bode plot for a system. Determine the value of K such that the phase margin is 45°.

Solution. Since

we have

The requirement that ω_c is the gain crossover

Note that ω_b/ω_n is a function of ζ .

Solution. The bandwidth ω_b is defined as the frequency at which the magnitude is -3 dB, we use -3.01 dB

Then

from which we get

Example Problems and

or

Figure 7–134 shows a

- A–7–18.** A Bode diagram of the system is shown in Figure 7–135. From the diagram, it can be determined that the damping ratio ζ is 0.5. Determine the damping ratio ζ . Also, determine the transfer function.

Solution. Referring to the Bode diagram, the damping ratio ζ and the natural frequency ω_n can be determined as follows:

Since, from Figure 7–10, $\angle G(j\omega) = 0^\circ$, the phase margin is unity. Also, the calculation of the transfer function $G(s)$ is

- A–7–19.** A closed-loop control system is shown in Figure 7–10. The stability criterion is to be satisfied. The transfer function of the plant element must be of the form

How can we obtain a stable closed-loop system? Suggest a possible transfer function of an unstable linear element.

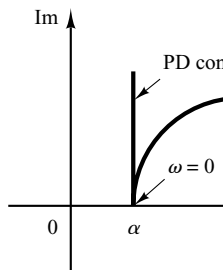
Solution. One possible transfer function of an unstable element by using the stability criterion is

Example Problems and Solutions

the phase angle of G
 $0 < \omega < \omega_1$ and acts

Figure 7-137

(a) Polar plots of a lead network and a proportional-plus-derivative controller;
 (b) polar plots of a lag network and a proportional-plus-integral controller.



(a)

Solution. The open-loop



Figure 7-138
Control system.

Example Problems and

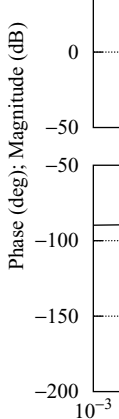


Figure 7-139

Bode diagram of

$$G(s) = \frac{10s + 1}{s(s + 0.5)(s + 1)}.$$

Figure 7-140

Nyquist plot of
 $G(j\omega)$ showing that
the phase margin is 60° .

Example Problems and

Figure 7-141
Bode diagram of
 $G(s)$ given by
Equation (7-32).

Figure 7-143

Bode diagram of
 $G(s)/[1 + G(s)]$,
where $G(s)$ is given
by Equation (7-32).

Example Problems and

With $K = 2$, the comp
 We shall next plot

Figure 7–144
 (a) Control system;
 (b) compensated
 system.

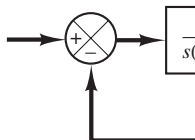
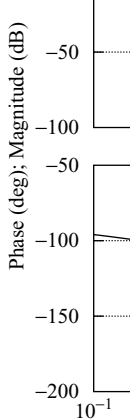


Figure 7-145
Bode diagram of
 $G_1(s)$.



Example Problems and

and

MATLAB Program 7-147 is shown in Figure 7-147.

MATLAB

<pre>numc = [1 0 0 0]; denc = [1 0 0 0]; w = logspace(-1, 1, 1000); bode(numc, denc, w); title('Bode Plot of the System')</pre>

Example Problems and Solutions

MATLAB Program 7
sated systems. The res
ed system exhibits a s
as follows:

Unit-Ramp Response
system. Since $K_v = 2$

Example Problems and

Figure 7–149

Unit-step responses
of the uncompensated
and compensated
systems.

MATLAB Program
response is obtained a
Figure 7–150. The com
inal uncompensated s

Figure 7–150

Unit-ramp responses
of the uncompensated
and compensated
systems.

0

A–7–25. Consider a unity-feed

Design a lag–lead con
phase margin is 50° , a

Example Problems and

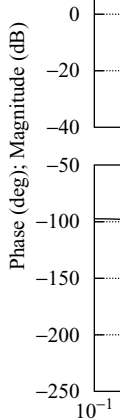


Figure 7-151

Bode diagram of

$$G(s) = 40/[s(s+1)(s+4)].$$

sator contributes -6 dB to the magnitude. To meet this requirement, it is necessary to choose a corner point (2 rad/sec, -6 dB) on the asymptotic approximation of this line and then choose the gain. In this consideration, the corner frequency is 2 rad/sec and $\omega = 4$ rad/sec. The transfer function becomes

Combining the transfer functions of the two blocks, the transfer function of the system becomes

Example Problems and Solutions

Using MATLAB Program 7-1, the transfer function $G_c(s)$ and polynomial den1 was

a =
b =
con
ans

we have

$$\frac{C(s)}{R(s)} =$$
$$=$$

To determine the den

Define

Example Problems and

Amplitude

0

0

0

0

Figure 7–154
Unit-step response
curve of the
compensated system.

Figure 7-155

Unit-ramp response
of the compensated
system.

B-7-1. Consider the unity-feedback system with
loop transfer function:

$$G(s) = \frac{10}{s + 1}$$

Problems

$$G(s) = \frac{s + 0.5}{s^3 + s^2 + 1}$$

This is a nonminimum-phase system. Two open-loop poles are located in the right-half plane. The following follows:

Open-loop poles at $s = -1.4656$

$$s = 0.2328 + j0.7$$

$$s = 0.2328 - j0.7$$

Plot the Bode diagram of $G(s)$ with MATLAB. The phase-angle curve starts from 0° and approaches -270° .

B-7-16. Consider the closed-loop system shown in Figure 7-157. $G(s)$ has no poles in the right-half s plane.

If the Nyquist plot of $G(s)$ is as shown in Figure 7-158(a), is this system stable?

If the Nyquist plot is as shown in Figure 7-158(b), is this system stable?

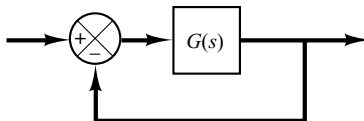


Figure 7-157
Closed-loop system.

Problems

B-7-21. Consider the system defined by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 6.5 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

There are four individual Nyquist plots involved in the system. Draw two Nyquist plots for the input u_1 and two Nyquist plots for the input u_2 . Write a MATLAB program to obtain the Nyquist diagrams.

Problems

ing of PID controllers has been developed and some new capabilities. Modified freedom PID controllers allow less switching (from commercially available

The usefulness of these systems. In particular, before analytical design in the field of process control schemes have proved many given situations

In this chapter we will discuss and Nichols tuning methods

resort to experimental
The process of s
ifications is known
PID controllers (m
responses or based
portional control ac
the following, are u
rules can, of course

Figure 8–1
PID control
of a plant.

Figure 8-3
S-shaped response
curve.



Thus, the PID contr

Second Method

proportional contr

K_{cr} at which the ou

hibit sustained oscil

apply.) Thus, the cri

Figure 8–4

Closed-loop system
with a proportional
controller.

Thus, the PID controller

Note that if the system is overdamped (i.e., $\zeta > 1$), then we can use the frequency of the sustained oscillation to determine the frequency of the sustained oscillation from the crossing point of the root-locus branches.

Section 8-2 / Ziegler-Nichols

Figure 8–6
PID-controlled
system.

The PID controller has
the control system with

Figure 8–7

Block diagram of the
system with PID
controller designed
by use of the
Ziegler–Nichols
tuning rule (second
method).

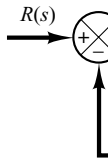


Figure 8–8

Unit-step response curve of PID-controlled system designed by use of the Ziegler–Nichols tuning rule (second method).

Figure 8–10

Unit-step response of
the system shown in
Figure 8–6 with PID
controller having
parameters

$$\begin{aligned}K_p &= 39.42, \\T_i &= 3.077, \text{ and} \\T_d &= 0.7692.\end{aligned}$$

Amplitude

double zero at $s = -0.1$. The two poles cancel each other. The dominant poles are at $s = -0.1 \pm j0.995$. On the other hand, when $\zeta = 0.5$, the zero is not quite dominant because it has a considerable effect on the system response. This is much larger than the effect of the poles. (In the latter case, the zero is much larger than the poles.)

It is possible to make the system response to this will take a lot of time. The computational approach is to use the computational approach to the computational approach. The maximum overshoot of the system is the problem obtained in the system.

8-3 DESIGN OF PID CONTROLLERS APPROACH

In this section we present a systematic approach.

Consider the system shown in Figure 8-3. We design a PID controller for a system whose phase is 50° or more, and whose response curves of type 1 or 2.

Let us choose the following parameters:

Section 8-3 / Design of PID Controllers

Figure 8–14
Bode diagram of
 $4/[s(s^2 + 1)]$.

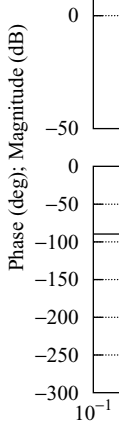


Figure 8–15

Bode diagram of

$$G(s) = 4(5s + 1) / [s(s^2 + 1)].$$

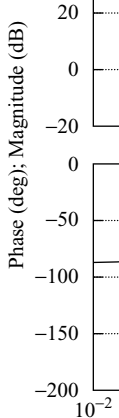
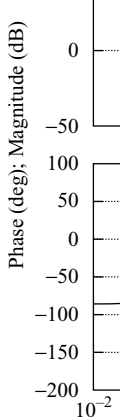


Figure 8–16

Bode diagram of

$$\frac{4(5s + 1)(0.25s + 1)}{[s(s^2 + 1)]}.$$



title
xla
yla
%*
nu
de
t =
c =
plo
titl
xla
yla
tex

Figure 8–18
Unit-ramp input and
the output curve.

gram begin with the re-
ing on the system and
from several seconds to

In this program the

will produce a table of
exhibit $m < 1.10$ —the

$R(s)$
→

Figure 8–19
PID-controlled
system.

Section 8–4 / Design

```
2.2000 0.5000  
2.2000 0.9000  
2.4000 0.5000  
2.4000 0.7000  
2.4000 0.9000  
2.6000 0.5000  
2.6000 0.7000  
2.8000 0.5000  
2.8000 0.7000  
3.0000 0.5000  
3.0000 0.7000  
sortsolution =
```

```

% greater than
K = sortsolution
K =
    2.8000
a = sortsolution
a =
    0.7000
gc = tf(K*[1 -2])
G = gc*g/(1 + s)
step(G,t)
grid % See Fig

```

Section 8-4 / Design

1
0.8
0.6
0.4
0.2
0

Amplitude

Figure 8–21

Unit-step response of
the system with
 $K = 2.8$ and $a = 0.7$.
(The maximum
overshoot is 0.24%.)

such that the closed-loop system is stable and has a desired response in the unit-step response. Assume that the search for the best controller is done by trial and error.

Determine the best controller for the system.

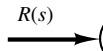


Figure 8–23
PID-controlled
system with a
simplified PID
controller.

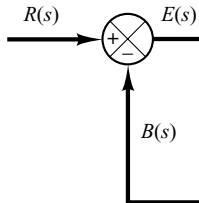
Section 8–4 / Design

	4.1000
	4.6000
	4.8000
	5.0000
	sortsolution =
	sortsolution =
	3.2000 0
	4.2000 0
	3.4000 0
	4.4000 0
	3.6000 0
	4.6000 0
	4.8000 0
	3.8000 0

Figure 8–24

Unit-step response
curve of the system
with $K = 3.2$ and
 $a = 0.9$.

Figure 8–25
(a) PID-controlled
system;
(b) equivalent block
diagram.



I-PD Control.

PID control and PI control are used to control a process with a step change in the reference. However, if the process has a non-zero steady-state gain, it may be advantageous to use a PD controller in the feedback path. Such a control scheme is called I-PD control.

Notice that the reference is not integrated in I-PD control, it is integrated in the feedback path. This is the main difference between I-PD control and I-P control.

Section 8-5 / Modified PID Control

actions to the reader. In the literature, PID schemes lie between the two extremes just considered. In these schemes, a third, derivative, controller is added to another controller in the feedback loop. The final two-degrees-of-freedom controller is a two-degrees-of-freedom control

8-6 TWO-DEGREES-OF-FREEDOM CONTROLLER

Consider the system shown in Figure 8-1. The disturbance input $D(s)$ is added to the reference input $R(s)$. The controller $G_c(s)$ is the transfer function of the controller. The plant $G_p(s)$ is the transfer function of the plant. We assume that the plant is a two-degrees-of-freedom control

Among the three cases
given, the remaining
a one-degree-of-freedom

Next consider the
of the plant. For this
respectively, by

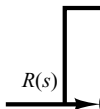


Figure 8–30
Two-degrees-of-
freedom control
system.



Figure 8–31
Two-degrees-of-
freedom control
system.

if the disturbance in

and assuming the sy

Requirement P

desired that the maximum difference between arbitrarily

where we choose the n smallest systems. The smaller the n , the more systems. In some cases, no combination of n systems allow a higher upper bound than the n smallest. At least one set of the n smallest systems instead of searching for a better set. Finding a reasonable reduction is not easy. To determine all closed

maximum overshoot

The response of the
can be made to ex
tion (8–4) is that it g
the settling time, the

The controller C



Figure 8–33
Two-degrees-
of-freedom control
system.

Note that the presence of the disturbance w adds a new term to the step disturbance d .

Let us assume that the disturbance w is given by



Figure 8–34
Search regions for
 a , b , and c .


```

den = [1
      y =
      m =
      s =
      s =
      ts =
      if m <
      k = k+
      table(
      end
      end
      end
      end
end

```

curves of the system with

is shown in Figure 8–7.
Using these values of

Using these closed-loop

$$s^2(s + 1) -$$

or

$$s^3 + (1 + 10K)$$

Section 8–7 / Zero-

Figure 8–35

(a) Response to unit-step reference input

($a = 4.2, b = 2, c = 12$);

(b) response to unit-step disturbance input

($a = 4.2, b = 2, c = 12$).

–0.0

By equating the coeff

we choose

Design Step 2: Next,

Section 8-7 / Zero-

Figure 8–36

(a) Response to unit-step
reference input

($a = 3.2, b = 2, c = 12$);

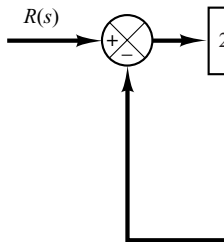
(b) response to unit-step
disturbance input

($a = 3.2, b = 2, c = 12$).

our problem becomes
step, ramp, and accele

Since the numer
[Although we want “s
zero steady-state erro

Figure 8–37
Block diagram of the
designed system.



input, (b) response to
unit-acceleration
reference input.

The response to the u
are shown in Figures
input and acceleration
Hence, the designed c

EXAMPLE 8-5

Consider the control s
design problem consi
plant transfer function

Design of $G_{c1}(s)$: Find the transfer function $G_{c1}(s)$ such that the system response to the step-disturbance $D(s)$ is zero. The block diagram which relates the output $Y(s)$ to the disturbance $D(s)$ is given by

Figure 8–40
Control system.

Figure 8–41

Root-locus plots of
 $5K(s + a)^2/[s(s + 1)(s + 5)]$ when $a = 3$,
 $a = 4$, $a = 4.5$, and
 $a = 6$.

Thus,

To check the response function $Y(s)/D(s)$.

0.08
0.07
0.06
 $y_d(t)$ 0.05
0.04
0.03
0.02
0.01
0

Figure 8–42
Response to unit-
step disturbance
input.

1.4
1.2
1
0.8
0.6
0.4
0.2
0

$y_r(t)$

Figure 8–43

(a) Response to unit-step reference input;
(b) response to unit-ramp reference input; (c) response to unit-acceleration reference input.

A-8-1. Describe briefly the PI controller.

Solution. The PI controller is a

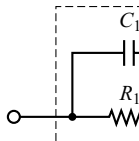
The PI controller is a controller that has the characteristic of zero steady-state error for a step change in the reference signal. The PI controller is a

Assume that the gain

$E(s)$
→

Figure 8–44
PID controller.

Example Problems and



$E_i(s)$

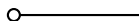


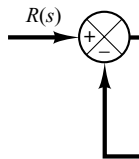
Figure 8–45
Modified PID
controller.

Note that such a d

control system and re

- A-8-5.** Consider the system s
sume that disturbance
erence input $R(s)$ is n
a very important cons

Figure 8-47
PID-controlled
system.



Example Problems and

With this PID control

For the reference

The response to a unit
The resulting response
imum overshoot is 7.3
characteristics.

Example Problems and

unit-step disturbance input; (b) response to unit-step reference input.

- A-8-6.** Consider the system s the dominant closed-loop poles. Choose $a = 1$ and the designed system.

Solution. Since

Figure 8–50
Root-locus plot of
the compensated
system.

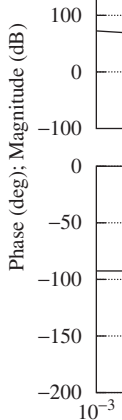
Example Problems and

where $\hat{G}_c(s)$ is to be 4 sec^{-1} , we have

$$K_v =$$

Figure 8–52
Control system.

Figure 8–53
 Bode diagram of
 $G(s) =$
 $40(s + 0.1)/[s(s^2 + 1)]$.



Example Problems and

Figure 8–54

Bode diagram of

$$G(s) = 40(s + 0.1) / [(0.1526s + 1) / s(s^2 + 1)].$$

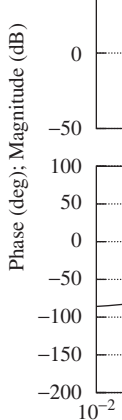


Figure 8–55

Unit-step response of
 $C(s)/R(s) = (6.104s^2 +$
 $40.6104s + 4)/(s^3 +$
 $6.104s^2 + 41.6104s + 4).$

Example Problems and

Nyquist Plot.

all in the left-half s plane. The Nyquist diagram here is not too complicated for stability analysis. For more complicated systems, it is not easy to count the

A MATLAB program
resulting Nyquist plot is

Example Problems and

Figure 8–59
Redrawn Nyquist
plot.

Using the Nyquist plot, the point by the Nyquist locus, the details near the -1

From this diagram
Hence, $N = -3$. Since
we have $P = 3$. Then,
the right-half s_1 plane

- A–8–8.** Show that the I-PD-co
system with input filter

The closed-loop transfer function is

- A-8-9.** The basic idea of the (phenomenon) within the feedback path, it is a PID control scheme.

Compare, qualitatively, the response of the system to the disturbance input.

Solution. Consider first the I-PD-controlled system. Since, in the I-PD control scheme, the derivative of the PID-controlled system is faster than the PID-controlled system.

Next, consider the I-PD-controlled system. In the I-PD-controlled system (problem A-8-8), the PID-controlled system, provided

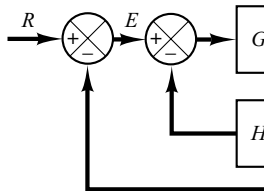
Example Problems and

Figure 8–61

(a) Block diagram of control system with input filter;
(b) modified block diagram.

Figure 8–62

(a) Control system;
(b) addition of the
internal feedback
loop to modify the
closed-loop
characteristic.



Example Problems and

A possible MATLAB
will satisfy the given

$R(s)$
→

Figure 8–63
Control system.

Example Problems and

Example Problems and

$$K = 27,$$

are two of the best choices. From these curves, we see that the maximum overshoot is more important than a

A-8-13. Consider the two-degree-of-freedom system given by

Assuming that the noise $n(t)$ is a white noise, the designed system satisfies

1. The response to the reference $r(t)$ is of the order of 1 (on the order of 1)

In what follows, we
us choose the dominant
closed-loop pole at s

The double zero at s
simple calculation, we

The controller $G_{c1}(s)$

The constant K must

Example Problems and

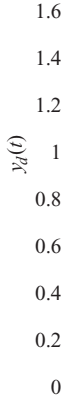


Figure 8–67
Response to unit-
step disturbance
input.

0.8
0.6
0.4
0.2
0

$y_r(t)$

Figure 8–68

(a) Response to unit-step reference input;
(b) response to unit-ramp reference input;
(c) response to unit-acceleration reference input.

Example Problems and

Figure 8–68
(continued)

overshoot in the unit-step response is 1.2 sec. The steady-state error is 0.05. Therefore, the designed controller is

Finally, we determine the

$G_c(s) = E_o(s)/E_i(s)$ is

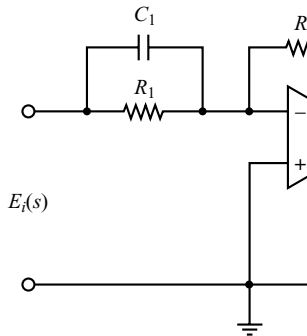


Figure 8–70

Electronic PID controller.

Problems

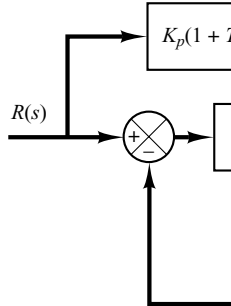


Figure 8–72

(a) PID-controlled system with feedforward control.

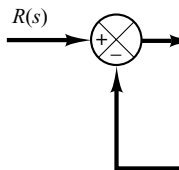


Figure 8–74
Control system.

Problems

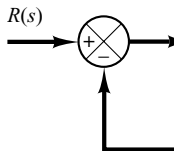


Figure 8–76

(a) Control system with a feedforward path. The feedforward path enters the summing junction with a negative sign.

(a), (b), (c) Two
degrees-of-freedom
systems.

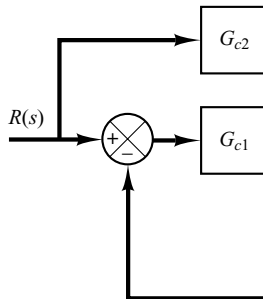


Figure 8–78
Three-degrees-of-
freedom system.

Problems

Figure 8–79
Control system.

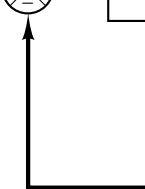


Figure 8–82

Two-degrees-of-freedom

Problems

terms of n first-order vector-matrix differential equations, the mathematical representation of the state variables, the complexity of the equations for output systems can be less complicated than those required for the equations.

This chapter and the next deal with systems. Basic material is presented in the next chapter.

* It is noted that in this book the **conjugate transpose** of matrix A is denoted by A^H (a matrix whose elements are the complex conjugates of the elements of A).

in what follows we
 Equation (9–1) or (9–2)
 diagonal (or Jordan)

Controllable Canonical Form
 a controllable canonical

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix}$$

transpose of that of

Diagonal Canon

tion (9–2). Here we
distinct roots. For th

The diagonal canon

$$\begin{bmatrix} \dot{x}_4 \\ \cdot \\ \cdot \\ \cdot \\ \dot{x}_n \end{bmatrix} =$$

$$y =$$

Section 9-2 / State

The characteristic e

The eigenvalues of

Diagonalization
eigenvalues is given

has the eigenvalues

will yield

This is in the Jordan

Section 9-2 / State

then, by substituting E

By premultiplying bo

or

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} =$$

Noting that the product $|\mathbf{P}^{-1} \mathbf{P}|$, we obtain

Thus, we have proved the desired transformation.

Nonuniqueness
The solution for the variables is not unique for

Section 9-2 / State

There are many (a
system. One possibl

Transformation
function from state

iu must be specified
has three inputs (u
implies u_2 , and 3 im
If the system ha

or

may be used. (See E

Section 9-3 / Trans


```
den =
```

```
1.0000 5.032
```

```
% ***** The same
```

```
[num,den] = ss2tf(
```

```
num =
```

```
0 -0.000
```

```
den =
```

```
1.0000 5.032
```

This is the MATLAB

Section 9-3 / Trans

Hence, the solution

$$x(i)$$

We shall now so

where $\mathbf{x} = n$ -vector

$$\mathbf{A} = n \times n \text{ c}$$

By analogy with the
power series in t , or

In terms of the mat

Since the matrix
systems, we shall ne

Matrix Expon
matrix **A**,

converges absolute
elements of e^{At} by u

Section 9–4 / Solvi

To prove this, note that

$$e^{(\mathbf{A}+\mathbf{B})t} = \mathbf{I} +$$

$$e^{\mathbf{A}t}e^{\mathbf{B}t} = \left(\mathbf{I} +$$

$$= \mathbf{I} +$$

$$+$$

Premultiplying both

The inverse Laplace

Note that

Hence, the inverse Laplace

$$\mathcal{L}^{-1}$$

Section 9–4 / Solving

motions of the system

If the eigenvalues are distinct, then the general solution is a linear combination of the n exponentials

In particular, if the eigenvalues are distinct, then the general solution is

$$\Phi(t) =$$

Since

the inverse of $(s\mathbf{I} - A)$

where $\mathbf{x} = n$ -vector

$\mathbf{u} = r$ -vector

$\mathbf{A} = n \times n$ c

$\mathbf{B} = n \times r$ c

By writing Equation

and premultiplying

c

The inverse Laplace
convolution integral

Solution in Terms of

If, however, the initial conditions
(9–40) must be modified

of vectors.

Cayley–Hamilton
proving theorems in
equations.

Consider an $n \times n$

The Cayley–Hamilton
equation, or that

To prove this theorem
That is,

adj

[See Problem A-9–

It is noted that the characteristic polynomial is given by the following procedure:

1. Form $\text{adj}(\lambda \mathbf{I} - \mathbf{A})$ and evaluate it in λ .
2. Determine $d(\lambda)$.
Choose the common denominator of the entries of $\text{adj}(\lambda \mathbf{I} - \mathbf{A})$ as a common denominator for the entries of $\text{adj}(\lambda \mathbf{I} - \mathbf{A})$.
3. The minimal polynomial is the lowest common denominator of the entries of $\text{adj}(\lambda \mathbf{I} - \mathbf{A})$.

Matrix Exponential

It is necessary to compute the matrix exponential. MATLAB provides the function `expm` to compute the matrix exponential.

Section 9-5 / Some

The inverse of matrix

Then it can be seen

S

Then, from Equation

$$e^{At}$$

Method 2. Since

we obtain

Section 9–5 / Some

$$\alpha_0(t)$$

If $\mathbf{A}$ is an $n \times n$ matrix, the number of roots determined is $m = n$. If $\mathbf{A}$ has only simple roots, ho

Case 2: Minimal polynomial
the case where the minimal polynomial has m roots and has other roots. The interpolation form is the determinant equation

roots will be apparent
to substitute the char
computations may,

EXAMPLE 9–8 Consider the matrix

Compute e^{At} using Sy
From Equation (9

Section 9–5 / Some

Linear Independence
independent if

where $c_1, c_2, \dots, c_n$ are

Conversely, the vectors
be expressed as a linear

9-6 CONTROLLABILITY

Controllability

if it is possible by means of an appropriate control function to transfer the system from any initial state $\mathbf{x}(t_0)$ to a final state $\mathbf{x}(t_f)$.

A system is said to be controllable if it is possible to determine this state by means of an appropriate control function.

The concepts of controllability and observability play an important role in the design of control systems. The conditions of controllability and observability are the solution to the control problem.

Section 9-6 / Control Systems

or

Referring to Equation

Substituting Equation

be of rank n , or con

is commonly called

EXAMPLE 9–10 Consider the system g

Since

the system is not com

Section 9–6 / Contr

Note that if the eigenvalue is λ , then λ^2 is also an eigenvalue. However, the converse is not true. If λ is a multiple eigenvalue of A , then λ^2 is also an eigenvalue of A^2 . If A is an eigenvalue of A^2 , then A is an eigenvalue of A^2 .

Let us define

Substituting Equation (9.10) into (9.9), we get

By defining

The square submatr

Suppose that we

If we define a new s

then substitution of

The condition for c
then be stated as fo

Section 9–6 / Contr

complete state control matrices.

It can be proved that controllability is that no cancellation occurs in the transfer function mode.

EXAMPLE 9-13 Consider the following

Clearly, cancellation occurs in the transfer function. (The system is not completely state

controllable if it is possible to transfer any given initial state $x(t_0)$ to any given final state $x(t_1)$ in a finite time $t_0 \leq t \leq t_1$.

It can be proved that a system is controllable if and only if the $m \times m$ matrix

is of rank m . (For a proof, see Equation (9-62) in the Appendix.)

Uncontrollable systems are physically disconnected from the control input.

Section 9-6 / Controllability

encountered with s
accessible for direct
the unmeasurable s
shown in Section 10
the system is compl

In discussing ob
Equations (9–63) an
by

then

to be n . (See Problem 9-6.)

From this analysis

The system described by (9-1) is

only if the $n \times nm$ matrix

is of rank n or has

observability matrix

Section 9-7 / Obse

where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix},$$

Note that the con
examine complete ob

Suppose that the tra

where **D** is a diagon

Then Equations (9–

Hence,

Section 9–7 / Obse

EXAMPLE 9-16 The following systems

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} - \\ \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \\ \cdots \\ 0 \end{bmatrix}$$

where $\mathbf{z}$ = state
 $\mathbf{v}$ = contr
 $\mathbf{n}$ = outpu
 $\mathbf{A}^*$ = conju
 $\mathbf{B}^*$ = conju
 $\mathbf{C}^*$ = conju

The principle of d
 (observable) if and
 To verify this pr
 complete state cont

Section 9-7 / Obse

A-9-1. Consider the transfer

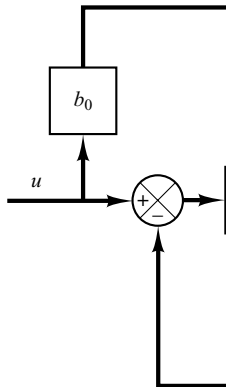
Derive the following
transfer-function system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix}$$

Then, clearly,

Example Problems and

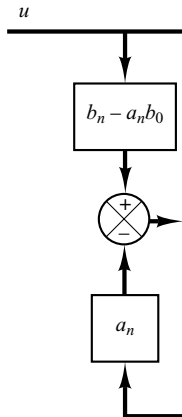
Figure 9-1
 Block diagram
 representation of the
 system defined by
 Equations (9-69)
 and (9-70)
 (controllable
 canonical form).



Now define state vari

Example Problems and

Figure 9–2
Block diagram
representation of the
system defined by
Equations (9–78)
and (9–79)
(observable
canonical form).



which may be rewritten

Example Problems and

Figure 9–3

Block diagram
representation of the
system defined by
Equations (9–84)
and (9–85) (diagonal
canonical form).

A-9-4. Consider the system of

where the system inv
 p_i 's being equal, the p
state-space representa

Example Problems and

Then, from the preceding
obtain

The state-space representation of the system in the Jordan canonical form is given by Equations (9–93) and (9–94).

A–9–5. Consider the transfer function

Obtain a state-space representation of the system.

Example Problems and Solutions

Let us write the i

where $\mathbf{w}$ is a vect
 $t = 0$. The solutio

(b) *Step response:* Le

Example Problems and

where $u(t)$ is the unit

Solution. For this sys

The state transition m

such that $\phi(\mathbf{A}) = \mathbf{0}$, c

The minimal polynomial of $\mathbf{A}$ is the monic polynomial of least degree such that $\phi(\mathbf{A}) = \mathbf{0}$.
matrix.

Let us suppose that $\mathbf{A}$ is a matrix of order n . Let $\phi(\lambda)$ be the characteristic polynomial of $\mathbf{A}$. Show that $\phi(\mathbf{A}) = \mathbf{0}$.
1, then the minimal polynomial of $\mathbf{A}$ is $\phi(\lambda)$.

Solution. By assumption $\mathbf{A}$ is a matrix of order n . Let $\phi(\lambda)$ be the characteristic polynomial of $\mathbf{A}$. Show that $\phi(\mathbf{A}) = \mathbf{0}$.

Example Problems and Solutions

Noting that $(\lambda \mathbf{I} - \mathbf{A})$

Since the greatest com

Therefore,

Then, from this last eq

A simple calculation gives

$$\mathbf{A}^3 - 5\mathbf{A}$$

$$= \begin{bmatrix} 8 & 4 \\ 0 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Example Problems and

Hence,

c

As a check, let us com

$$\phi(\mathbf{B}) = \mathbf{B}^2 - 3\mathbf{B} +$$

For the given matrix $\mathbf{B}$
acteristic polynomial.
in a Jordan chain, the

Clearly, there is no c
Consequently, the min

Thus, the minimal pol
Since the character

we obtain

Example Problems and

If a square matrix can
and it can be obtained

Then

where **D** is a diagonal

Hence,

we obtain

Note that $e^{\mathbf{J}t}$ is a tr
may be) the principal
example, if matrix $\mathbf{J}$ h

then

Example Problems and

Then the polynomial

$$f(\lambda) =$$

=

takes on the values $f(\lambda_1), f(\lambda_2), \dots, f(\lambda_m)$ at the points $\lambda_1, \lambda_2, \dots, \lambda_m$. This is the *interpolation formula*.

data $f(\lambda_1), f(\lambda_2), \dots, f(\lambda_m)$.
 $f(\lambda_1), f(\lambda_2), \dots, f(\lambda_m)$

Any other representation of the polynomial $f(\lambda)$.

Equations (9–102) and (9–103) are
for example, $(\lambda \mathbf{I} - \mathbf{A})^{-1} = \frac{1}{\lambda^2 - 2\lambda + 1} (\lambda \mathbf{I} - \mathbf{A})$

Show that Equation (9–102) is valid
that $m = 4$.

Example Problems and Solutions

$$\begin{vmatrix} \lambda_i & \lambda_j & \lambda_k \\ \lambda_i^2 & \lambda_j^2 & \lambda_k^2 \\ \lambda_i^3 & \lambda_j^3 & \lambda_k^3 \end{vmatrix}$$

we obtain

$$\begin{aligned} \Delta &= f(\mathbf{A}) \\ &\quad - f(\lambda_i \mathbf{I}) \\ &\quad + f(\lambda_j \mathbf{I}) \\ &\quad - f(\lambda_k \mathbf{I}) \\ &\quad + f(\lambda_l \mathbf{I}) \\ &= \mathbf{0} \end{aligned}$$

Solution. Since the minimal polynomial $\phi(\lambda)$ can be written as

An arbitrary function

where the minimal polynomial has degree $m - 1$ or less. Hence

where $\alpha(\lambda)$ is a polynomial

Example Problems and

Solution. Referring to the previous problem, λ_1 and λ_2 are the same for this A .

Note that $\lambda_1 = \lambda_2 = -1$. If the initial value problem is e^{At} , we have

where $\alpha_0(t)$, $\alpha_1(t)$, and

Example Problems and

$$\mathbf{A} = n \times n \text{ m}$$

$$\mathbf{B} = n \times r \text{ m}$$

$$\mathbf{C} = m \times n \text{ m}$$

is completely output c

is of rank m . (Notice
complete output cont

Solution. Suppose that
the initial output, can
 $0 \leq t \leq T$. That is,

$$e^{\mathbf{A}\tau} =$$

and $\mathbf{B}_j$ is the j th column

From this last equation
 $p-1; j=1,2,\dots,r)$

Q

is m , then so is the rank
 (where $p \leq h \leq n -$

Example Problems and

Notice that cancellations occur in the transfer function. Because the system has a pole at $s = 1$, this is an unstable system. There are many systems

A-9-17. A state-space representation

by the fact that the ori
to Equation (2–29), fo

If we use Equations (

[Note that the same tr
Clearly, cancellation o

Example Problems and

Hence, we obtain, for

Notice that from Equ

$e^$

where $m(m \leq n)$ is th

$$\mathbf{C}e^{\mathbf{A}t}\mathbf{x}(0) =$$

which means that

which implies that rank $\mathbf{C} = n$.
Equation (9–126), we obtain

and $\mathbf{x}(0)$ can be determined.

Hence, we have proved
that $\mathbf{x}(0)$ and $\mathbf{y}(t)$ are

$$\mathbf{y}(t) = \mathbf{C}e^{At}\mathbf{x}(0)$$

Example Problems and

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} =$$

Obtain the transfer function $Y(s)/U(s)$.

B-9-5. Consider the following matrix $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Obtain the eigenvalues $\lambda_1, \lambda_2, \lambda_3$, and λ_4 of the

Then obtain a transformation matrix $\mathbf{P}$ such that

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \text{diag}(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$$

$$\begin{bmatrix} \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Is the system completely state controllable and observable? Is the system completely output c

Problems

We begin by presenting the basic concepts of systems and control systems. Then we discuss the quadratic optimal control problem and its solution. Finally, we discuss the quadratic optimal control problem in the context of robust control systems.

Outline of the Course

Chapter 10–2 discusses the basic concepts of systems and control systems with the derivation of the state-space representation. Chapter 10–3 presents the basic concepts of the state-space representation. Chapter 10–4 discusses the design of the state feedback controller. Chapter 10–5 presents state observer design. Chapter 10–6 presents the design of the state observer. Also, transfer function models are presented. Chapter 10–7 presents the design of regulator systems.

where $\mathbf{x}(0)$ is the initial
response characteri

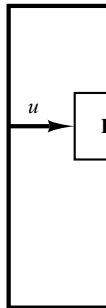


Figure 10–1
Closed-loop control
system with
 $u = -\mathbf{K}\mathbf{x}$.

$$|s\mathbf{I} -$$

where $\mathbf{I}_q$ is a q -dimensional identity matrix.

Section 10-2 / Pole

state controllable),
modified to

where

This characteristic equation is defined by Equation 10-10, seen as follows: Since

the characteristic equation

$$|s\mathbf{I} - \mathbf{A} + \mathbf{B}\mathbf{K}| = 0$$

Section 10-2 / Pole

Solving the preceding
we obtain

$$\begin{aligned}\mathbf{K} &= [\delta \\ &= [\alpha\end{aligned}$$

Thus, if the system
placed by choosing

We have thus p
placement is that th

It is noted that i
then it is possible to
desired locations fo
are stable. So the en

(10–13), rewritten to

$$\mathbf{K} = [\mathbf{a}$$

Determination

is of low order ($n \leq 4$), the characteristic polynomial may be equated to the matrix $\mathbf{K}$ as

Substitute this $\mathbf{K}$ matrix into the characteristic equation to equate it to $(s - \mu_1)(s - \mu_2) \dots (s - \mu_n) = 0$.

Equation, we have

$$\phi$$

We shall utilize Eq. (10.10) in the following derivation, we consider the following derivation

Consider the fol

$$\mathbf{I}$$

$$\tilde{\mathbf{A}}$$

$$\tilde{\mathbf{A}}^2$$

$$\tilde{\mathbf{A}}^3$$

$$\begin{bmatrix} \mathbf{B} \\ \mathbf{C} \end{bmatrix}$$

Premultiplying both

$$\begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{B} \\ \mathbf{C} \end{bmatrix} = \mathbf{A}$$

which can be rewritten

This last equation gives

For an arbitrary

$$\mathbf{K} =$$

Section 10-2 / Pole

(We make such a choice that the resulting matrix $\mathbf{K}$ will result in a reasonable matrix $\mathbf{K}$.)

Figure 10–2
Regulator system.

Method 2: By defini

and equating $|s\mathbf{I} - \mathbf{A}|$

$$|s\mathbf{I} - \mathbf{A} +$$

Section 10–2 / Pole

As a matter of course
With this state feedback

It is noted that if
mended, since all ma
hand computations b
equation with unknow

command place rec
loop poles.

For single-input
multiple-input system

It is noted that v
tional problem may
place command is p
sired set of closed-l

To use the com
program:

where **J** matrix is th

Section 10–3 / Solv

where t is the time du

Then obtain x_1 , x_2 , and

and use the plot com
response curves are sh

Section 10-3 / Solv

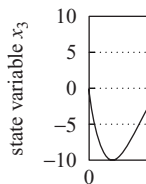
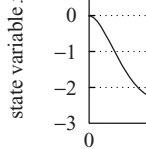


Figure 10–3
Response to initial
condition.

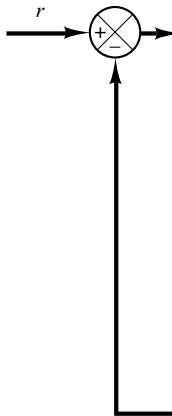


Figure 10–4
Type 1 servo system
when the plant has
an integrator.

Then Equation (10-

Equation (10–25) d

The design of th
totically stable regu
tion $\mathbf{e}(0)$. If the sys
then, by specifying t
 $\mathbf{K}$ can be determine

The steady-state
($t = \infty$), we have, fr

where

The state-feedback gain K is determined by the MATLAB Program 10–4.

Section 10–4 / Design of a State-Feedback Controller

```

AA = [0
BB = [0;
CC = [1
DD = [0

% *****

t = 0:0.0
y = step(
plot(t,y)
grid
title('Uni
xlabel('t
ylabel('C

```

Design of Type

has no integrator (the system is to insert an integrator and the plant, as shown in the form of the type 1 system we obtain

where $\mathbf{x}$ = state vector

Section 10-4 / Design

$$\begin{bmatrix} \dot{\mathbf{x}}(\infty) \\ \dot{\xi}(\infty) \end{bmatrix}$$

We shall design an a constant values, res

Notice that at st

$$\begin{bmatrix} \dot{\mathbf{x}}(\infty) \\ \dot{\xi}(\infty) \end{bmatrix}$$

Noting that $r(t)$ is subtracting Equation

$$\begin{bmatrix} \dot{\mathbf{x}}(t) - \dot{\mathbf{x}}(\infty) \\ \dot{\xi}(t) - \dot{\xi}(\infty) \end{bmatrix}$$

The state error
Equation (10–40):

If the desired eigenvalues are
specified as $\mu_1, \mu_2, \dots, \mu_n$, the
gain constant k_I can be determined
from Eq. 10–2, provided that the system is
controllable. Note that

has rank $n + 1$, the system is
controllable. (See Problem 10–1.)

Section 10–4 / Design of a



Figure 10–8
Inverted-pendulum
control system.

Let us define the state

Then, referring to Equation (10-1), we can write x as the output of the

Matrix $\mathbf{P}$ is given by

The rank of this matrix is completely state dependent. 10–6 produces the state

Program 10-7 may be used to obtain the unit impulse response that, to obtain the unit impulse response

Figure 10-10 shows the unit impulse response $(= \xi)$ versus t . Notice that the response is approximately 4.5 s

or

Section 10-4 / Des

```
title('x')
xlabel('x')

subplot(2,2,2)
title('x')
xlabel('x')

subplot(2,2,3)
title('x')
xlabel('x')

subplot(2,2,4)
title('x')
xlabel('x')
```

Hence, for $r = 1$, we have

It is noted that, as $\epsilon \rightarrow 0$,
then we must modify the
simulations must be re-

10-5 STATE OBSERVERS

In the pole-placement
state variables are a
available for feed

Section 10-5 / State

or observation error
output. The initial en
state. Thus, we defin

where $\tilde{\mathbf{x}}$ is the estim
are the output y an
matrix, is a weighti
the measured output
the model output an
block diagram of th

Then Equation (10-

From Equation (10-
mined by the eigen
the error vector wil
converge to $\mathbf{x}(t)$ reg
 $\mathbf{A} - \mathbf{K}_e \mathbf{C}$ are chos
asymptotically stabl
origin) with an adeq

If the plant is c
choose matrix $\mathbf{K}_e$ s
observer gain matr
shall discuss this ma

Section 10-5 / Sta

Comparing the characteristic polynomial $|s\mathbf{I} - (\mathbf{A} - \mathbf{K}_e\mathbf{C})|$ with that $\mathbf{K}_e$ and $\mathbf{K}^*$ are

Thus, using the matrix inversion lemma, the observer gain $\mathbf{K}_e$ is determined by the relationship $\mathbf{K}_e = \mathbf{K}^* \mathbf{C}^{-1}$.

Necessary and Sufficient Conditions
 necessary and sufficient conditions for the desired eigenvalues to be assignable are

where $\mathbf{K}_e$ is an $n \times$

and

[Refer to Problem A

Section 10-5 / Sta

the preceding Acke

$$\mathbf{K} = [0$$

As stated earlier, th
Equation (10–64). T

$$\mathbf{K}_e = \mathbf{K}^*$$

system performance
performance. In ma
a compromise betw

EXAMPLE 10–6 Consider the system

where

We use the observed s

Section 10–5 / Sta

the characteristic equation

Define

Then the characteristic equation

$$\begin{vmatrix} s & 1 \\ 0 & s \end{vmatrix} = s^2 + 1$$

higher, methods 1 and 2 are more suitable for a computer, while method 3 is more suitable for problems involving unknown parameters.

Effects of the
pole-placement design on the system response with feedback. In practice, the designer often has to need to design an observer as well as the controller. Figure 10–12. The design of the observer is being the determination of the observer poles, the characteristic equation and the selection of the observer gain to yield the desired observer response.

Let us now investigate the effect of the observer on the actual state $\mathbf{x}(t)$.

Section 10–5 / State Estimation

defined as the error

Substitution of the

Note that the obser

Combining Equatio

Then the transfer function

Figure 10–13 shows the transfer function

acts as a controller

Section 10–5 / State

Suppose that we use
control, or

where we choose the

Obtain the observ
control system. Then c
draw another block di
path. Finally, obtain th

Since

Equation (10-78) becomes

or

$$\begin{aligned}\begin{bmatrix} \ddot{\tilde{x}}_1 \\ \ddot{\tilde{x}}_2 \end{bmatrix} &= \begin{bmatrix} 2 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} -7 \\ -9 \end{bmatrix}\end{aligned}$$

The block diagram of

Section 10-5 / State

As a matter of course
MATLAB Program 10
shows a block diagram

The system, as a whole

$$|s\mathbf{I} - \mathbf{A}|$$

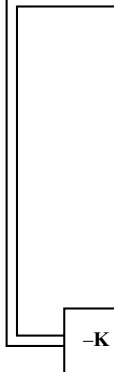
The characteristic equation is
Figure 10–14(b). Since

$$\frac{Y(s)}{R(s)}$$

Section 10–5 / State

```
e1 = [0  
e2 = [0  
  
subplot(2  
title('Res  
ylabel('st  
  
subplot(2  
title('Res  
ylabel('st  
  
subplot(2  
xlabel('t  
  
subplot(2  
xlabel('t
```

Figure 10–16
Observed-state
feedback control
system with a
minimum-order
observer.



or

The terms on the left-hand side of Equation (10.10) act as the output error, e , in the left-hand side of Equation (10.9). The measurable quantities y and u are

From Equation (10.10), the error e becomes

Noting that terms A and B are the dynamics of the unmeasured

Full-O
K_e

where $\mathbf{v}$ is a row vector

then we can write $\hat{\mathbf{x}}$

This equation gives

Figure 10–17 shows
with the minimum-cost
and $u = -\mathbf{K}\tilde{\mathbf{x}}$.

Next we shall de
(10–86) can be mod

$$\dot{\tilde{\mathbf{x}}}_b$$

This is the error equation for the full-order observer.

The error dynamics for the full-order observer are

is $n - 1$. (This is the order of the observer.)

Section 10-5 / State Feedback

Note that $u_1, u_2, \dots$
equation

$|s\mathbf{I}$

Also, if Ackermann
modified to

where L is the vector
order observer, we

provided L does not
the transpose.] For
commands:

or

Section 10-5 / Sta

Since

$$\tilde{\mathbf{x}} = \begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix}$$

we have

Noting that

$$\mathbf{A}_{bb} =$$

the equation for the n

$$\begin{bmatrix} \hat{r} \\ \hat{r} \end{bmatrix}$$

or

Section 10–5 / Sta

$[-90 \ -29 \ -4]$

Figure 10–18

System with observed state feedback
Example 10–8.

Equations (10–106)
By considering u as

Since the input to the
of the observer con

This transfer function

Section 10–5 / Sta

Referring to Section

$r =$

Figure 10–19
Regulator system.

```

j = [-11] 2 -11]
K = acker(A,B,J)

K =

    1.2500    1.2500

% Obtaining the
Aaa = 0; Aab = [
L = [-10 -10];
Ke = acker(Abb',
Ke =

    10
   -24

```

Section 10-6 / Des

```

Ahat = A
Bhat = B
Fhat = F
Atilde = A
Btilde = B
Ctilde = C
Dtilde = D
[num,d] = [num,d]
num =
    9.1000
den =
    1.0000

```

Second trial: Let us
but modify the obser

Thus,

Using MATLAB, w

Section 10–6 / Des

Notice that this is a
System 2. We shall
condition:

By substituting $u =$

$$\dot{\mathbf{x}} = \mathbf{Ax} -$$

$$= \mathbf{Ax} -$$

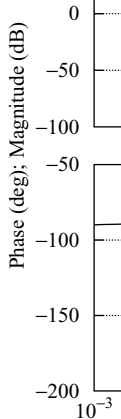


Figure 10–22

Bode diagram for the open-loop transfer function of System 2.

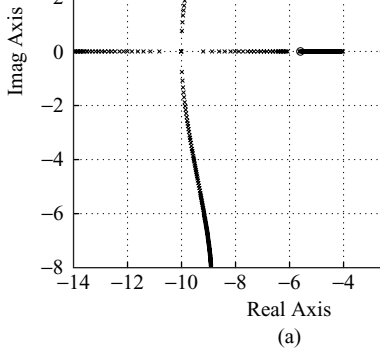


Figure 10–24

(a) Root-locus plot of the system with observer poles at $s = -4$ and system with observer poles at $s = -4$

In this section we consider placement-with-observer controller. The block diagram is shown in Figure 10–25. If the system has a relative degree r , then it is conceivable, each has a relative degree r in Figures 10–26 (a) and (b).

r

Figure 10–25
Regulator system.

we get

Figure 10–27

Control system with
observer controller
in the feedforward
path.



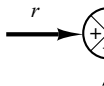
The transfer function is
Program 10-17. The

$$G_c(s)$$

Figure 10–28
Regulator system
with observer
controller.



Figure 10–29
Control system with
observer controller
in the feedforward
path.



The unit-step response
maximum overshoot is v



Figure 10–31

Control system with
observer controller
in the feedback path.

Output y

0.8

0.6

0.4

0.2

0

Figure 10–33

The unit-step response of the control system designed by the pole placement approach without observer. (The closed-loop poles are at $s = -1 \pm j$, $s = -8$.)

estimate the u

3. In designing a desired closed-loop transfer function, compared, and
4. The bandwidth of the observer pole and the frequency noise
5. Adding an observer. In some cases, an observer means that the system is unstable. Then the controller is unstable. Then
6. When the system is unstable, it is advisable to c

importance of the error signal e is that the control vector u is chosen

As will be seen later, the optimal control law. Therefore, the goal is to minimize the performance index J over the initial state $\mathbf{x}(0)$. The block diagram of the optimal regulator system is shown in Figure 10-35.

Figure 10-35
Optimal regulator
system.

$$J = \int_0^\infty \mathbf{x}^*(\mathbf{Q}$$

Since all eigenvalues of $\mathbf{A}$ have negative real parts, $\mathbf{x}(\infty) \rightarrow \mathbf{0}$. Therefore,

Thus, the performance index J is a function of $\mathbf{P}$ and $\mathbf{P}$.

To obtain the optimal gain $\mathbf{K}$, it follows: Since $\mathbf{R}$ is a positive definite matrix, we can write

matrix $\mathbf{A} - \mathbf{BK}$ is s

Finally, note that
rather than the stat

then the index can b

to

and the design step
matrix $\mathbf{K}$.

Section 10–8 / Qua

This equation can be written as

$$\begin{bmatrix} 0 & 0 \\ p_{11} & p_{12} \end{bmatrix}$$

Figure 10–36
Control system.

Note that the control
under the given perfor
Since the character

if $\mu = 1$, the two close

These correspond to t

Solving Quadra
the command

Section 10–8 / Qua

Show that the system

whatever matrix $\mathbf{K}$ is

Define

Then

[K,P,E] = lo

Warning: M

K =

NaN NaN

P =

-Inf -Inf

-Inf -Inf

E =

-2.0000

-1.4142

Section 10-8 / Qua

MATLAB

```
% -----
```

```
A = [0 1
```

```
B = [0;1]
```

```
Q = [1 0
```

```
R = [1];
```

```
K = lqr(A
```

```
K =
```

```
1.0000
```

$$[K,P,E] =$$

$$K =$$

$$0.014$$

$$P =$$

$$4.26$$

$$2.49$$

$$0.014$$

$$E =$$

$$-5.095$$

$$-1.985$$

$$-1.985$$

Section 10-8 / Qua

EXAMPLE 10–13 Consider the system equations:

where

$$\mathbf{A} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The control signal u is

$$u = k$$

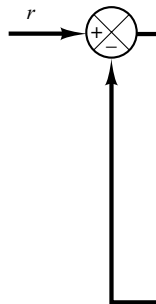


Figure 10–39
Control system.

and the output equation

To obtain the unit

where

$AA =$

MATLAB Program 1
shows the response cu

Figure 10–40

Response curves x_1
versus t , x_2 versus t ,
and x_3 versus t .

(10–114) is first
linear function
the determina
varying matrix

10–9 ROBUST CONTROL SYSTEMS

Suppose that given
sign a control system
mathematical model
model may be nonlinear
be difficult to analyze
system that will apply

ferences or errors b

Uncertain elements
tured uncertainty an
any parametric var
of the plant transfer
dependent uncertain
eling plant dynamic
may include a finite
included in the mod
certainty occurs in t
and its model is line

In this section w
tion we assume that
multiple uncertain c

and limit the magnitude
a small γ and require



Figure 10–41

Bode diagram and
the H_∞ norm $\|\Phi\|_\infty$.

where γ_m and γ_a are

EXAMPLE 10-14

Consider a control system. The stability and robust property will be discussed in

Robust Stability

$\tilde{G}$ = true plant d

G = model of pl

Δ_m = unstructur

We assume that Δ_m is related by

Then Inequality (10–1

Clearly, for a stable y ,
 $K(s) = 0$ is not the d
fer function for $K(s)$,
have robust performa
another reasonable
performance.

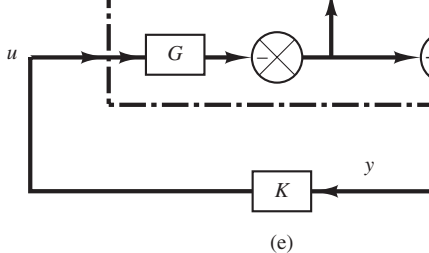


Figure 10-43

- (a) Block diagram of a system with unstructured
- (b)–(d) successive modifications of the block
- (e) block diagram showing a generalized plant
- (f) generalized plant diagram.

Our problem then becomes finding on the chosen W_m may be no $K(s)$ that satisfies (10–127) is called a minimum

Figure 10–45(a) is the best robust performance) are

Figure 10–44
Closed-loop system.

able. $z(s)$ is the controller $K(s)$. The equation for $u(s)$ of the general

The equation that r

Define the transfer disturbance $w(s)$ as

Section 10–9 / Rob

Hence,

EXAMPLE 10–15

Let us determine the ΔH° for the reaction in Example 10–14. We will use the ΔH°_f values in Table 10–1. Rewriting Inequality

then we obtain

which is the same as Q

Section 10–9 / Rob

(10–155) we are able to design a controller for a generalized plant diagram as shown in Figure 10–47.

***H* Infinity Control**

We will now discuss how to satisfy various stability and performance requirements for a generalized plant.

As mentioned earlier, the H_2 norm of the plant and weighted H_2 norm are measures of the plant's performance. Reference tracking and disturbance rejection control problems are solved by minimizing the H_2 norm of the transfer function from the disturbance to the error. The H_∞ norm is a measure of the plant's robustness to a specified value of the H_2 norm.

scope of the senior
graduate-level con-
ject in detail.

A-10-1. Consider the system of

Suppose that this system
matrix is less than n , or

Example Problems and

Since we have q linear
theorem to express ve

$$\left[\begin{array}{c} a \\ \vdots \\ a \end{array} \right]$$

Then Equation (10–11)

$$|$$

Example Problems and

where

A-10-2. Consider a completely

Define the controllab

Hence, the left-hand side of (1) is equal to $\sum_{i=1}^n a_i$, thus shown that Equation (1) holds. $\square$

The preceding derivation shows that

A-10-3. Consider a completely reducible representation ρ of a group G on a vector space V . Define

Define

Example Problems and Solutions

Hence, Equation (10-

Therefore, we need to

The left-hand side of

where

The rank of the contr

is 2. Thus, the system is
controllable canonical

Solution. Since

Example Problems and

which is in the control

A-10-5. Consider a system de

where

Hence,

Referring to Equation

Thus, the control sign

Example Problems and

(Note that, for the purposes of this problem, the matrix $\mathbf{K}$.)

Two MATLAB programs are provided: Programs 10–24 and 10–25.

Equation (10–147) may be written

We shall show that Eq. (10–148) may be written

Example Problems and

and

where the a 's are coefficients

Define also

$$(\mathbf{WN}^*)\mathbf{A}$$

Hence, we need to show

or

Example Problems and

Next define

Then Equation (10–13)

and Equation (10–152)

Referring to Equation

Hence,

γ

Thus, we have shown

Note that what we have shown is that n is an integer.

A-10-9. Consider a system defined by

Example Problems and

Then the state equation

or

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} =$$
$$=$$

The output equation is

$$\mathbf{K} =$$

where

For the original system

Hence, matrix $\mathbf{T}$ can a

Since $\mathbf{W} = \mathbf{W}^*$, we ha

and

Example Problems and

Combining Equations

we obtain

Equation (10–166) de
minimum-order obser

or

Since matrix

is of rank $n + 1$, the left
side of Equation (10–

$$\begin{bmatrix} \cdot \\ - \end{bmatrix}$$

Example Problems and

Figure 10–49
Mechanical system.



$\text{Abb} = [$
 $L = [-15$
 $\text{Ke} = \text{pl}$
 place: r
 $\text{Ke} =$
 14.4
 0.3

Response to Initial Co
 initial condition. Since

Example Problems and

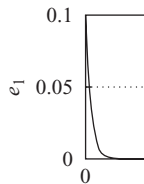
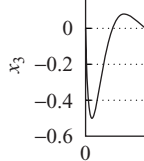


Figure 10–50
Response curves to
initial condition.

for the plant. Assume

Assume also that the

(a) $s = -8, s = -8$ for

(b) $s = -8$ for the m

Compare the resp

(a) for the full-order

x

$r = 0$



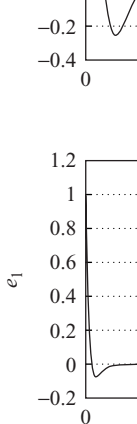
Figure 10–51
Regulator system.

Example Problems and

Now we find the m in (10–70), we have

This equation defines t . Program 10–29 produces the results. The results are shown in Figure 10–10.

Figure 10–52
Response curves to
initial condition.



Example Problems and

The response of the s
tained as follows: By

```

sys = ss(
t = 0:0.01:10
x = init
x1 = [1
x2 = [0
e = [0 0

subplot(2,1,1)
xlabel('x1')

subplot(2,1,2)
xlabel('x2')

subplot(2,1,3)
xlabel('e')

```

Example Problems and

```

Ke = 6;
Ahat = Abb - Ke*Aab;
Bhat = Ahat*Ke + Aba -
Fhat = Bb - Ke*Ba;
Atilde = Ahat - Fhat*Kb;
Btilde = Bhat - Fhat*(Ka
Ctilde = -Kb;
Dtilde = -(Ka + Kb*Ke);
[num,den] = ss2tf(Atilde
num =
    7  32
den =
    1  10

```

where $\mathbf{Q}$ is a positive-

Solution. The matrix

has the solution

Integrating both sides

Example Problems and

A-10-16. Consider the control system

where

Assuming the linear controller

determine the constant

is minimized, where t

Since in this system $\mathbf{Q}$

Since the system invol
Then Equation (10–17)

$$\begin{bmatrix} 0 & - \\ 1 & -A \end{bmatrix}$$

Example Problems and

The designed system is
under the assumed in

- A-10-17.** Consider the same inv
in Figure 10-8, where
shown in Figure 10-9.

Using MATLAB
performance index J

Example Problems and

[Refer to Equation (10–177)]

Equation (10–178) can be written as

The output equation is

xlabel
subplot
xlabel
subplot
xlabel
subplot
xlabel

Comparing the step responses, we notice that the response of the optimal regulator approaches the response of the optimal regulator with overshoot in the position.

Example Problems and

Cart P
0.4
0.2
0
-0.2

Figure 10–56
Cart position versus t
curve.

- A–10–18.** Consider the stability
10–57(a). Define
 $\tilde{G}$ = true plant dy
 G = model of pla
 Δ_a = unstructure

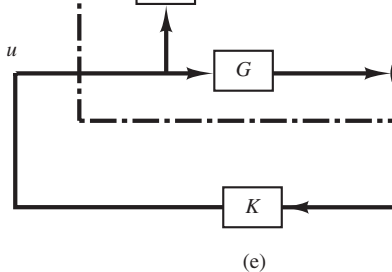


Figure 10-57

- (a) Block diagram of a system with unstructured
- (b)–(d) successive modifications of the block d
- (e) block diagram showing a generalized plant
- (f) generalized plant diagram.

Example Problems and

plant diagram for the
 Note that for this
 nous disturbance w is

Noting that $u(s) = K$
 of the P matrix as foll

To make this $\Phi(s)$ eq
 $P_{22} = -G$. Then, the

have the closed-loop poles at $s = -2 \pm j4$, $s =$
mine the state-feedback gain matrix $\mathbf{K}$.

B-10-4. Solve Problem B-10-3 with MATLAB

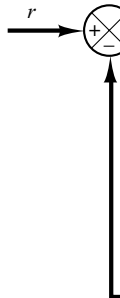


Figure 10-58
Type 1 servo system.

Problems

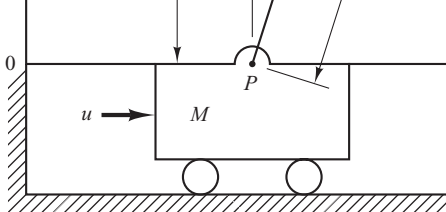


Figure 10–59

Inverted-pendulum system.

B–10–10. Consider the system defined by

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$$

$$y = \mathbf{C}\mathbf{x}$$

design observer controllers (one with a full-order the other with a minimum-order observer) for shown in Figure 10–60. The desired closed-loop pole-placement part are

$$s = -1 + j2, \quad s = -1 - j2, \quad s =$$

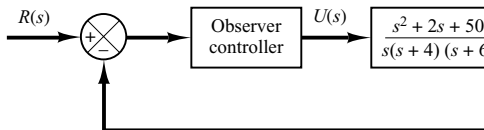


Figure 10–60

Control system with observer controller in the feedforward path.

Problems

such that the following performance index is m

$$J = \int_0^{\infty} (\mathbf{x}^T \mathbf{x} + u^2) dt$$

B-10-20. Consider the system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

It is desired to find the optimal control signal $u(t)$ that minimizes the performance index

$$J = \int_0^{\infty} (\mathbf{x}^T \mathbf{Q} \mathbf{x} + u^2) dt, \quad \mathbf{Q} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

is minimized. Determine the optimal signal $u(t)$.

where σ is the real part of s

Complex Function
an imaginary part of s

where G_x and G_y are the real and imaginary parts of $G(s)$. The angle θ of $G(s)$ is taken with respect to the positive real axis. The complex conjugate of $G(s)$ is denoted by $G^*(s)$.

Complex function
single-valued function of a complex variable

are satisfied, then the functions $G(s)$ and $H(s)$ are known as the Cauchy integrals of $G(s)$ and $H(s)$, respectively. The function $G(s)$ is analytic in the region $\text{Re}(s) > \sigma_0$. As an example,

Then

singular points. Singular points at infinity are called *poles at infinity*. Poles at finite points are called *zeros*.

If $G(s)$ approaches

has a finite, nonzero limit, the pole is called a *simple pole*. If the limit is infinite, the pole is called a *multiple pole*, or *higher-order pole*. A pole of order three is called a *third-order pole*, and so on.

To illustrate, consider

Appendix A / Laplace Transform

where c , the abscissa of convergence, is the real part of all singular points of $F(s)$ and is displaced by the real parts of all singular points.

Evaluating the inverse Laplace transform integral for finding $f(t)$ is discussed in Appendix B.

In what follows we discuss the most commonly encountered Laplace transform pairs and transforms.

13	
14	
15	
16	
17	

Appendix A / Lapl

27	
28	
29	
30	$\overline{\omega_2^2}$
31	

13	
14	
15	
16	
17	
18	

Appendix A / Lapl

distinct poles only, then

$$F(s) = \frac{B(s)}{A(s)}$$

where a_k ($k = 1, 2, \dots$) are the roots of $A(s)$ at $s = -p_k$. The value of a_k is determined by $(s + p_k)$ and let

$$\left[(s + p_k) \frac{B(s)}{A(s)} \right]_{s=-p_k}$$

EXAMPLE B-2 Obtain the inverse Laplace transform of

Here, since the degree of the numerator is less than the degree of the denominator, we must divide

Partial-Fraction
cussing the general
fraction expansion o
Consider the fol

The partial-fraction

Appendix B / Partial

the ratio of two polynomials.

The partial-fraction decomposition of

$$\frac{B(s)}{A(s)}$$

Comparing Equation (B-1) with

$$p(n) = -p_n; r(1) =$$

EXAMPLE B-4 Consider the following

Note that if $p(j) =$
pole $p(j)$ is a pole of n

For details, see Example

It is the MATLAB re

Note that the direct to

Appendix B / Partial

its sign.

2. If any row or a column is zero.
 3. If the element in a row (or another row) is zero.
 4. If, to any row or column is added, the value of the determinant is not changed.
 5. If a determinant is multiplied by a scalar, the value of the determinant is multiplied by the scalar.
- $n \times n$ matrix

Rank of Matrix
 matrix **M** of **A** such
 $r \times r$ submatrix (w
 As an example,

Appendix C / Vecto

Adjoint Matrix.

A_{ji} is called the adjoint

That is, the adjoint of $\mathbf{A}$ is the matrix $\mathbf{A}^*$ of $\mathbf{A}$, or

then the adjoint of A

Appendix C / Vectors

Note that

There are several
 $\mathbf{B} = n \times m$ matrix,
[$\mathbf{A}$

then the inverse of

Appendix C / Vectors

$$\int \mathbf{A}(t) \, dt = \left(\int \right)$$

Differentiation
can be differentiated

Here again the mul
al, not commutative

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